

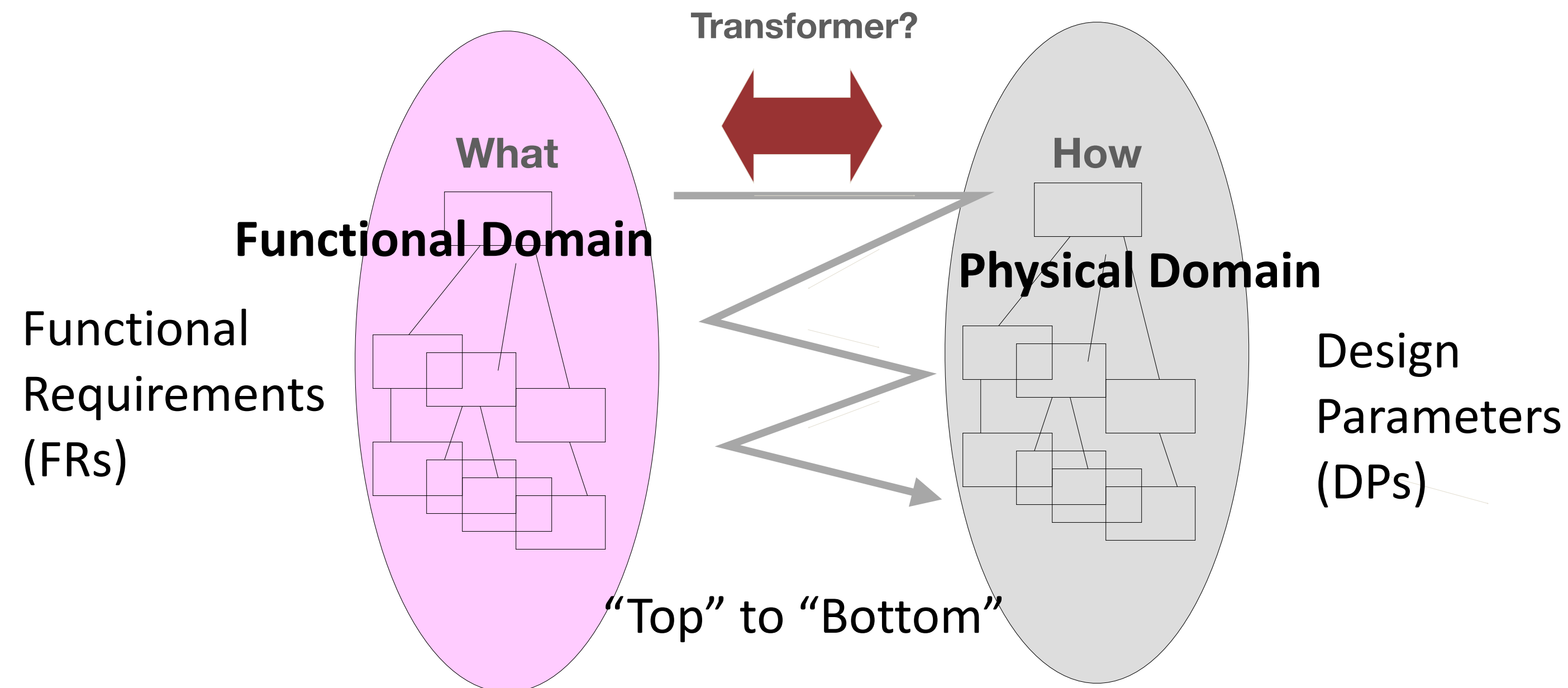
A close-up, low-angle shot of a rocket engine's nozzle and combustion chamber. The engine is firing, with bright blue flames and white smoke billowing out from the nozzle. The background is dark, suggesting a night sky or a dark environment. The engine's structure is metallic and complex, with various pipes and components visible.

AI for Design and Manufacturing Part 2: Thinking Design

Prof. Sang-Gook Kim
MIT

Design Principles

Search in the dark



$$\overrightarrow{\mathbf{FR}} = [\mathbf{DM}] \overrightarrow{\mathbf{DP}} \quad A_{ij} = \frac{\partial(FR_i)}{\partial(DP_j)}$$

Axiomatic Design

- What is a good design?
- Concept of Domains
- Functional thinking
- Top-down, zig-zagging decomposition
- Design axioms

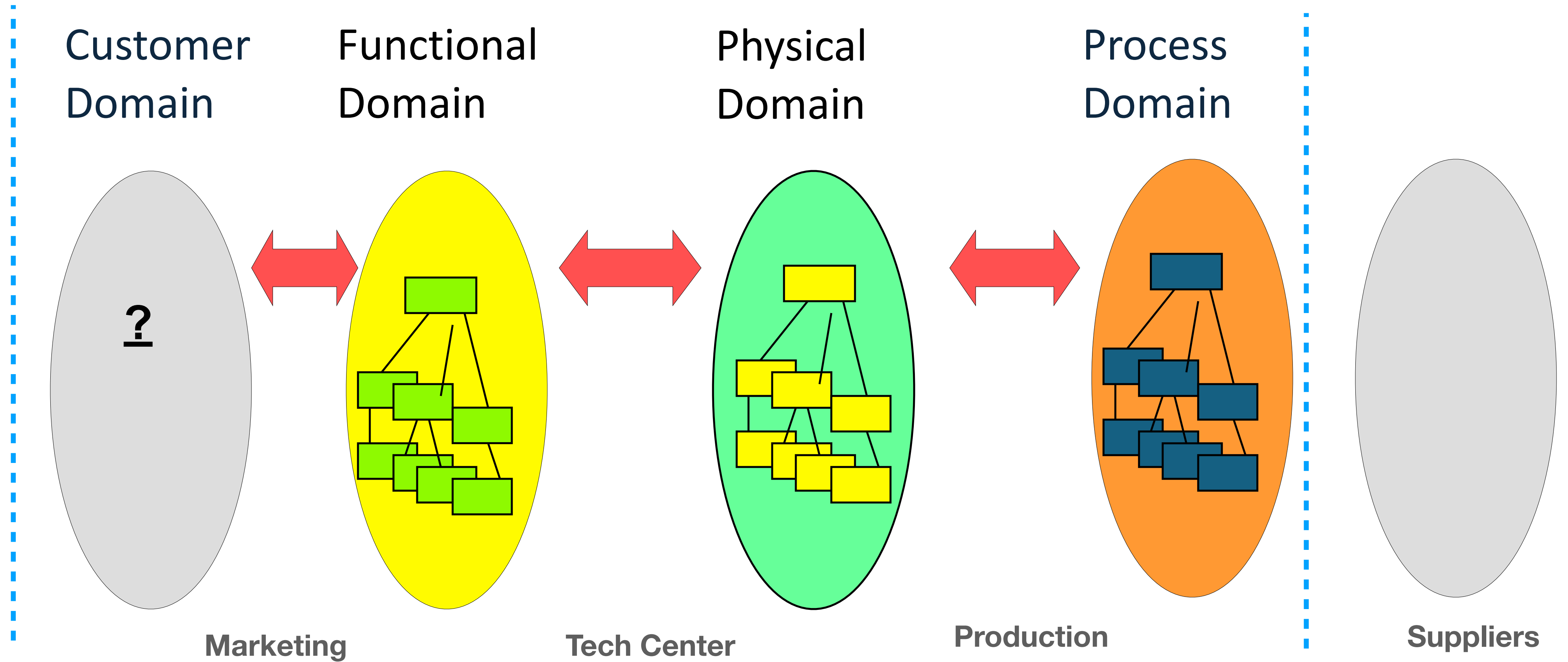
Nam Suh, MIT faculty (1970-2008), "Optimization of Manufacturing Systems through Axiomatics," CIRP Annals 1978

* S.G. Kim, et al. "Axiomatic design: Making the abstract concrete," 26th CIRP Design Conference, Procedia CIRP Volume 50, 2016

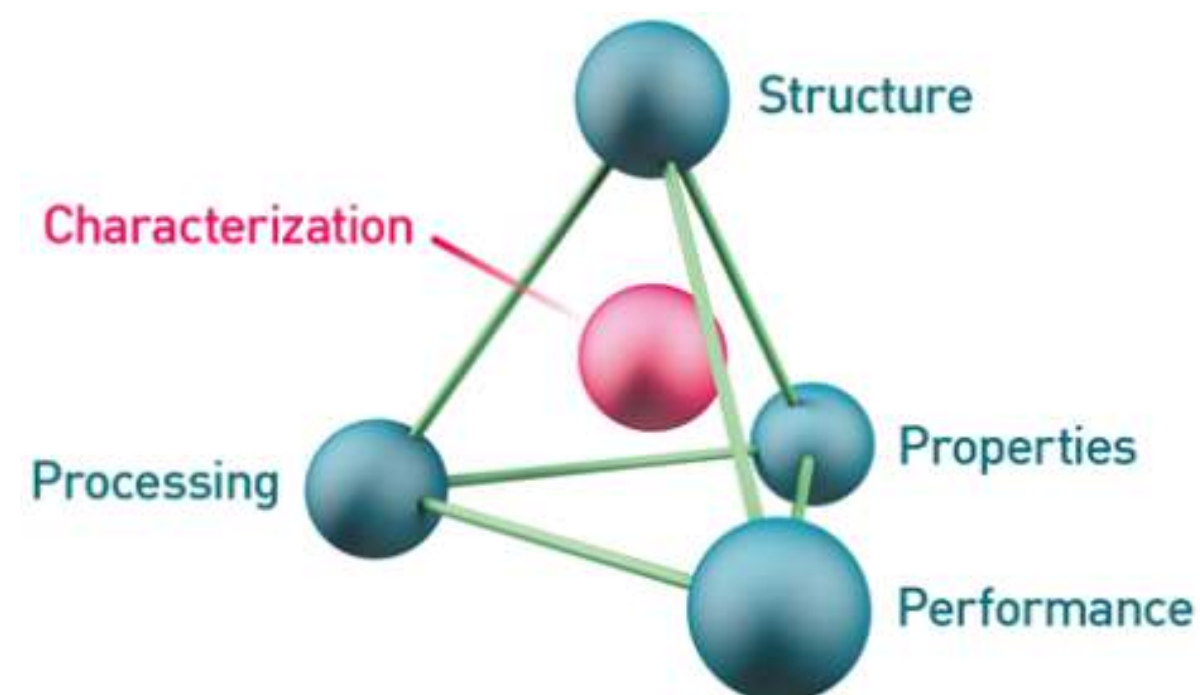
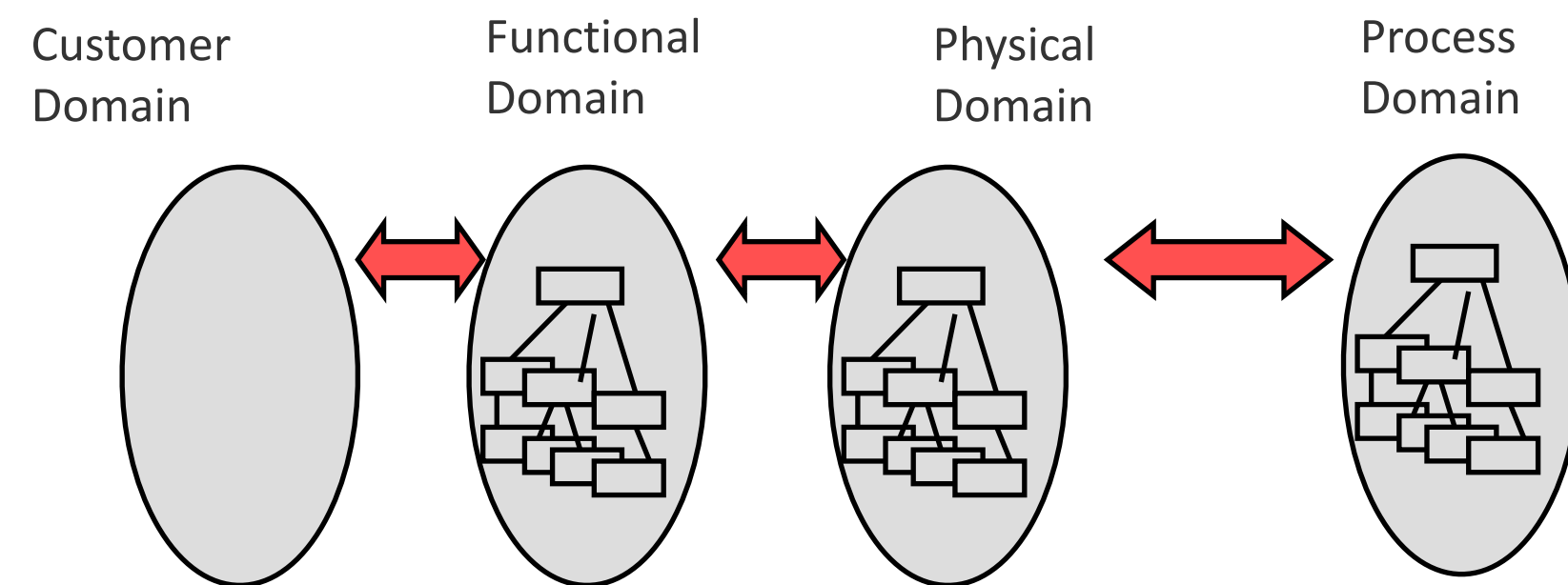
* S.G. Kim (EDs.), Advances in Multidisciplinary Engineering, ASME Press, NY, 2015

* S.G. Kim, Axiomatic Design, Encyclopedia of Production Engineering, DOI 10.1007/978-3-642-20617-7, # CIRP 2019, pp108

Four Design Domains



Four Domains of Systems



Material Science Tetrahedron

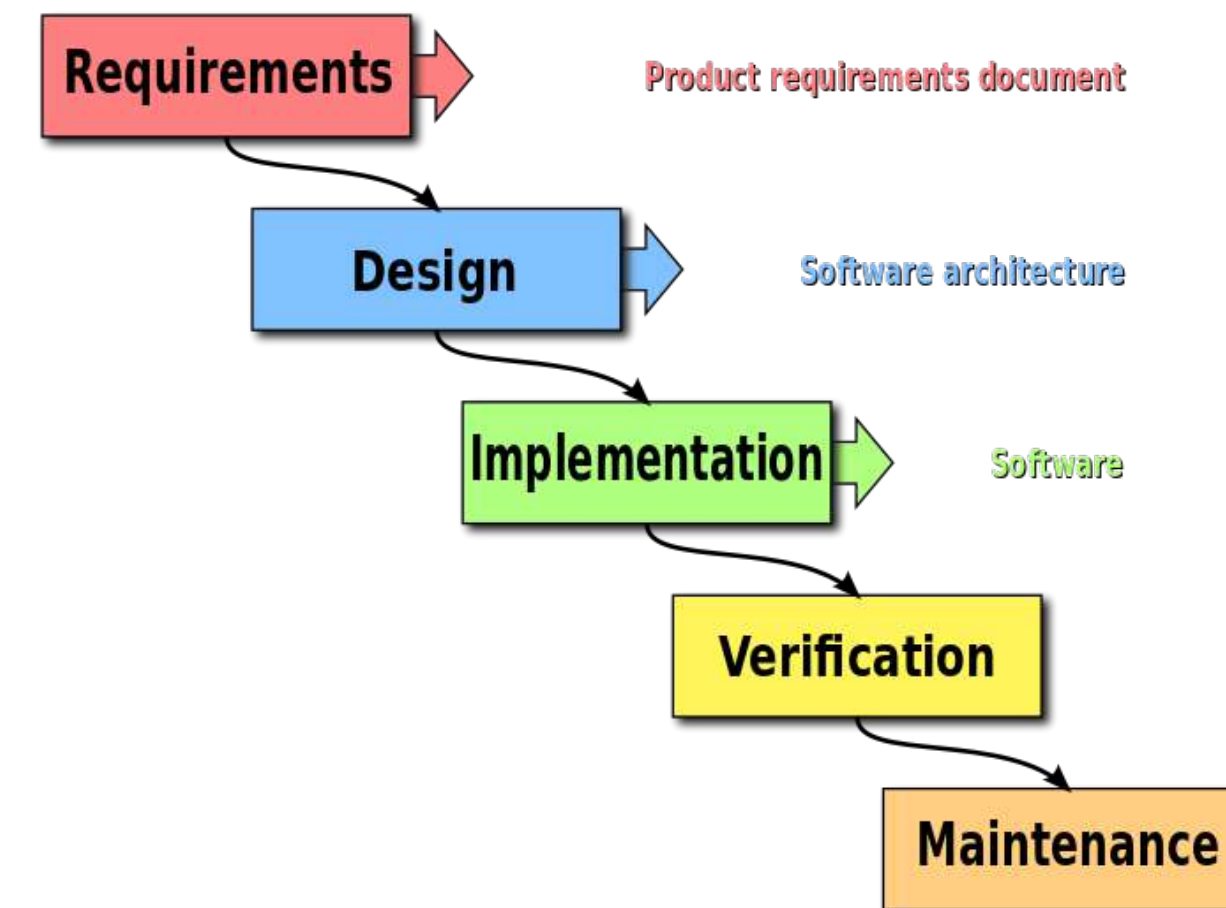
	Customer domain	Functional domain	Physical domain	Process domain
Manufacturing/Engineering	Customers' attributes	Functional requirements	Design parameters	Process variables
Materials	Performance	Properties	Microstructure	Processes
Software	Attributes	Output	Input variables	Subroutines
Organization	Customer satisfaction	Functions of organization	Programs & Offices	Resources, people
Business	ROI	Business goals	Business structure	Resources, human, capital

Designers' (Problem solvers') common mistakes

1. Not to realize concept of design domains, “What” to “How”
2. Not to realize that design process involves mapping from “What” to “How”
3. Physical thinking (not **solution neutral**)
4. Decomposition without zig-zagging domains (concepts)
5. Trying to make functionally coupled designs work.

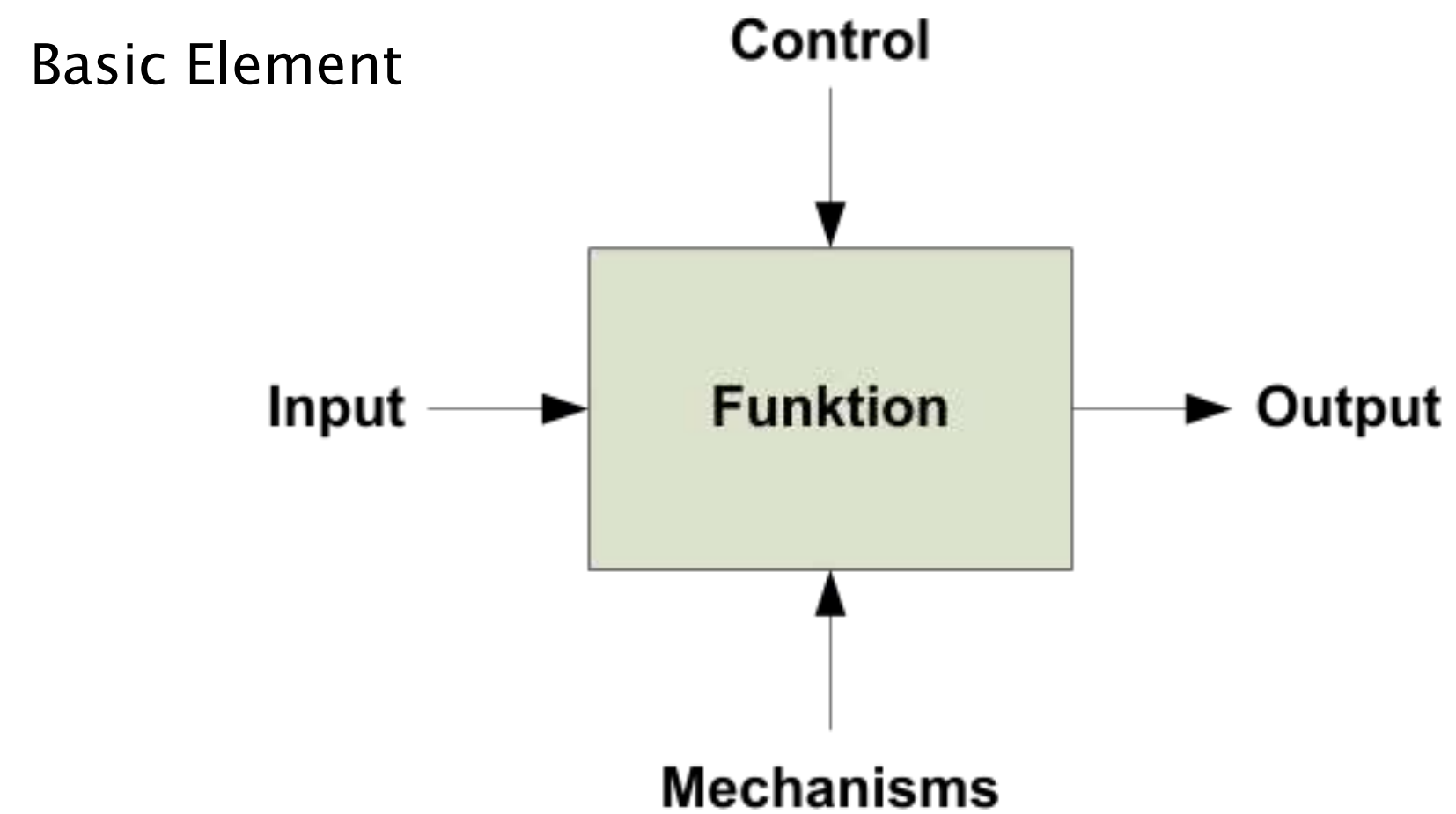
System Design of Software ?

- Waterfall development
- Prototyping
- Incremental development
- Rapid Application Development
- Spiral Development
- Agile Development
- SADT
- Productivity: \$/line, time to delivery, hard maintenance

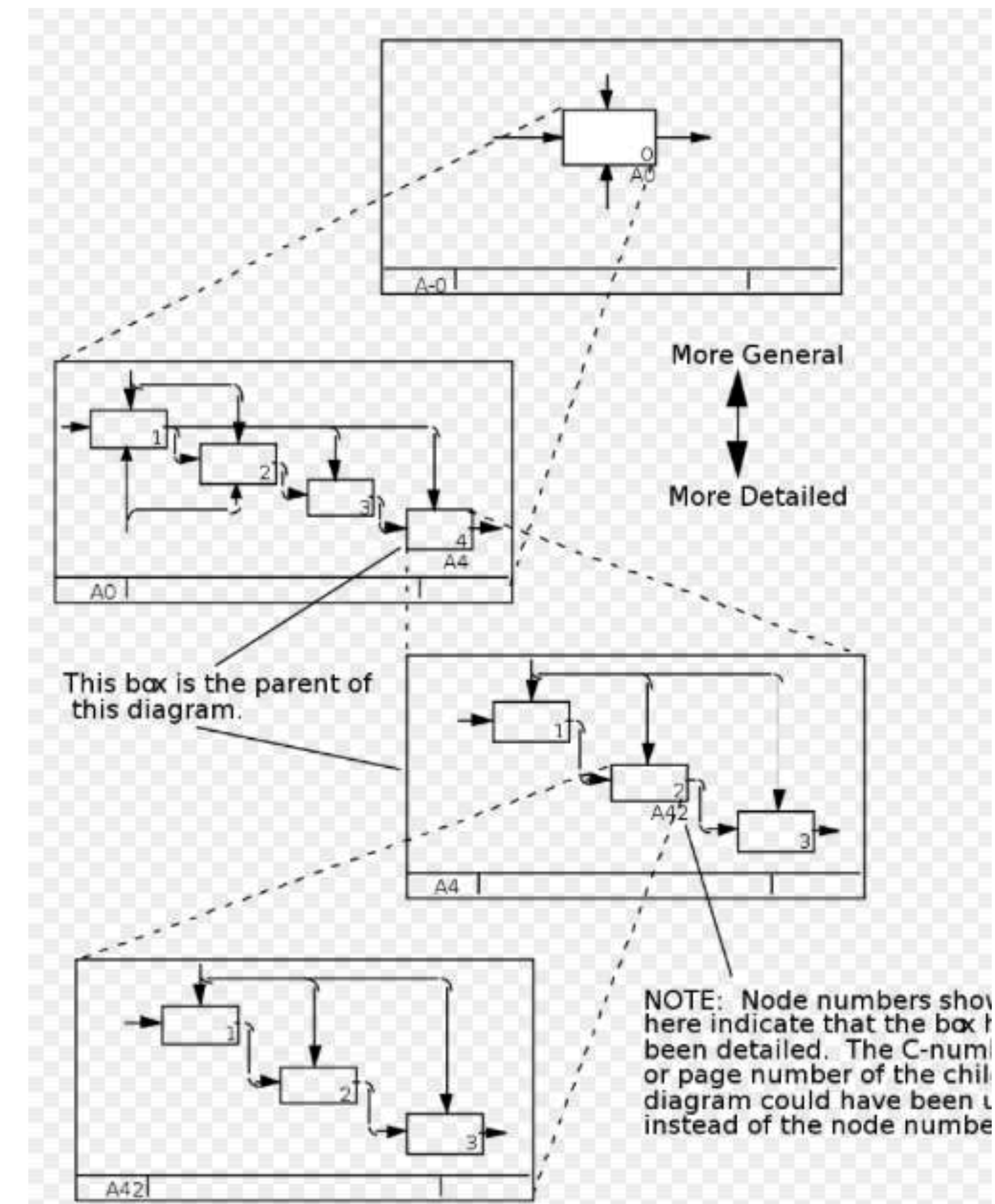


System Design of Software

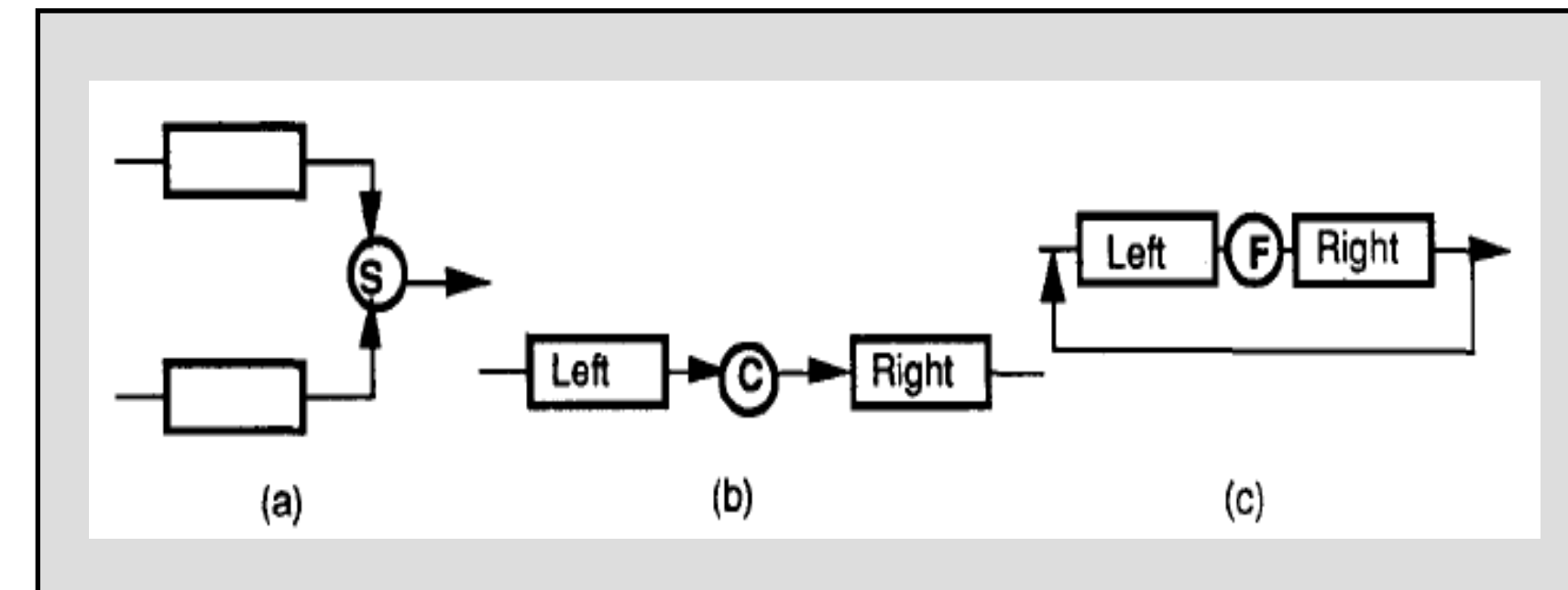
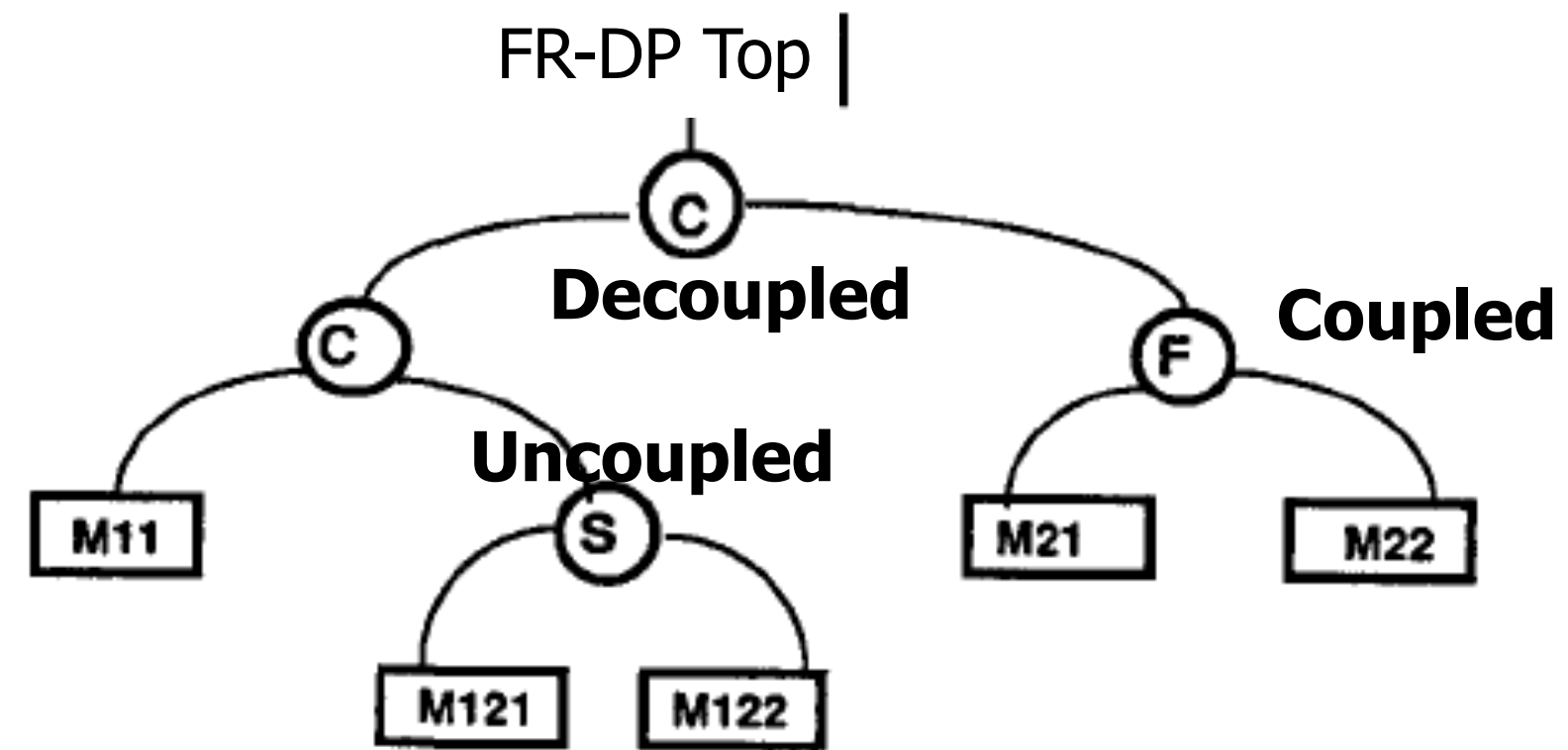
Structured Analysis and Design Technique (SADT)



https://en.wikipedia.org/wiki/Structured_analysis_and_design_technique



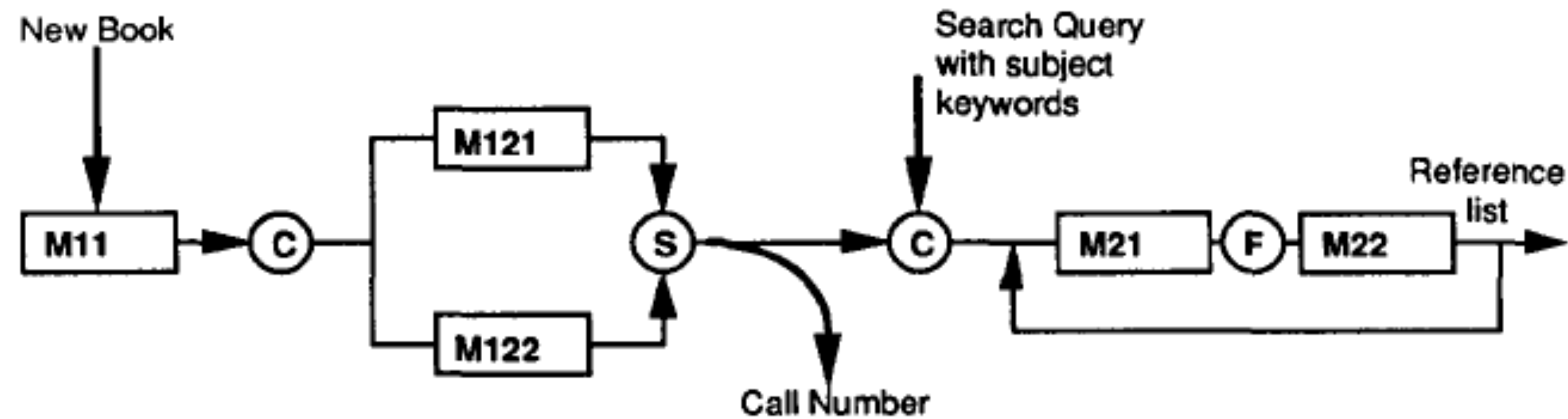
Module Junction Structure in Data Flow Domain



Uncoupled

Decoupled

Coupled



Why do we teach/learn systems thinking? (2.778)



Functional Thinking

Why do we teach/learn systems thinking? (2.778)



What would you do?

The U.S. airport with the longest average wait time at security is more than double the shortest. Photo: Marius Dobilas / Shutterstock.com.

Could we improve the passengers experience through the airport security?



- **Bottle necks mitigation**
- Screening success rate

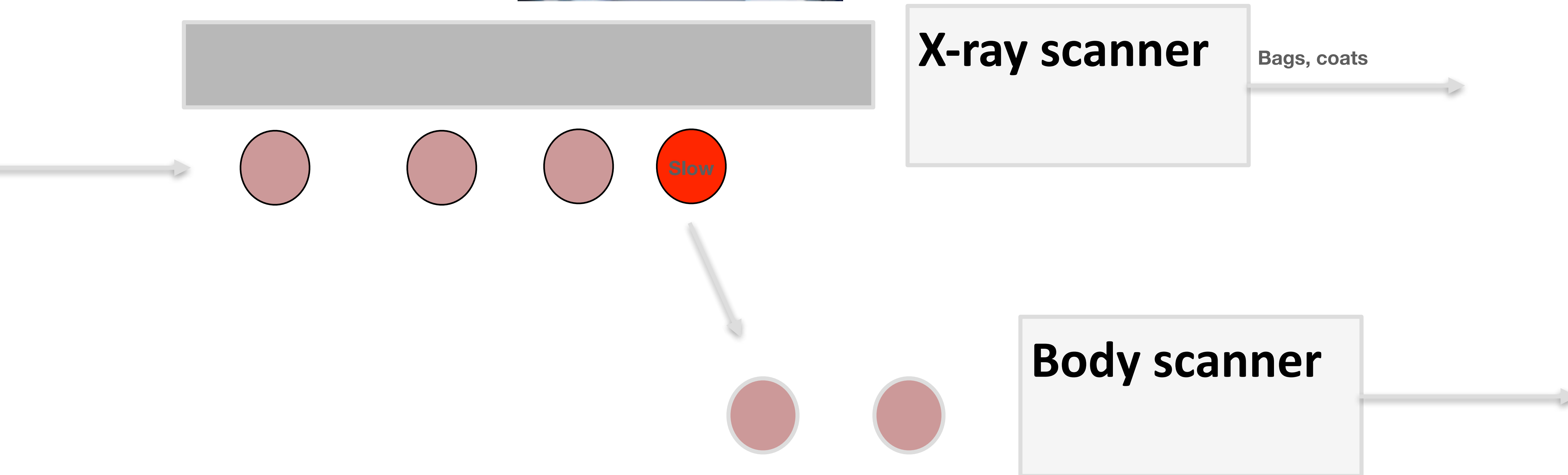
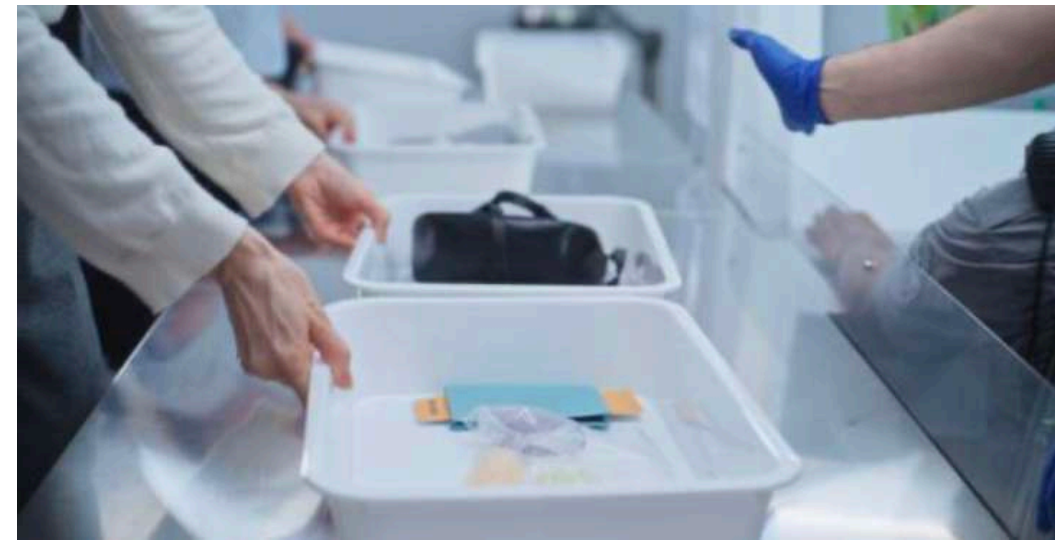
Could we improve the passengers experience through the airport security?

2 min thinking exercise

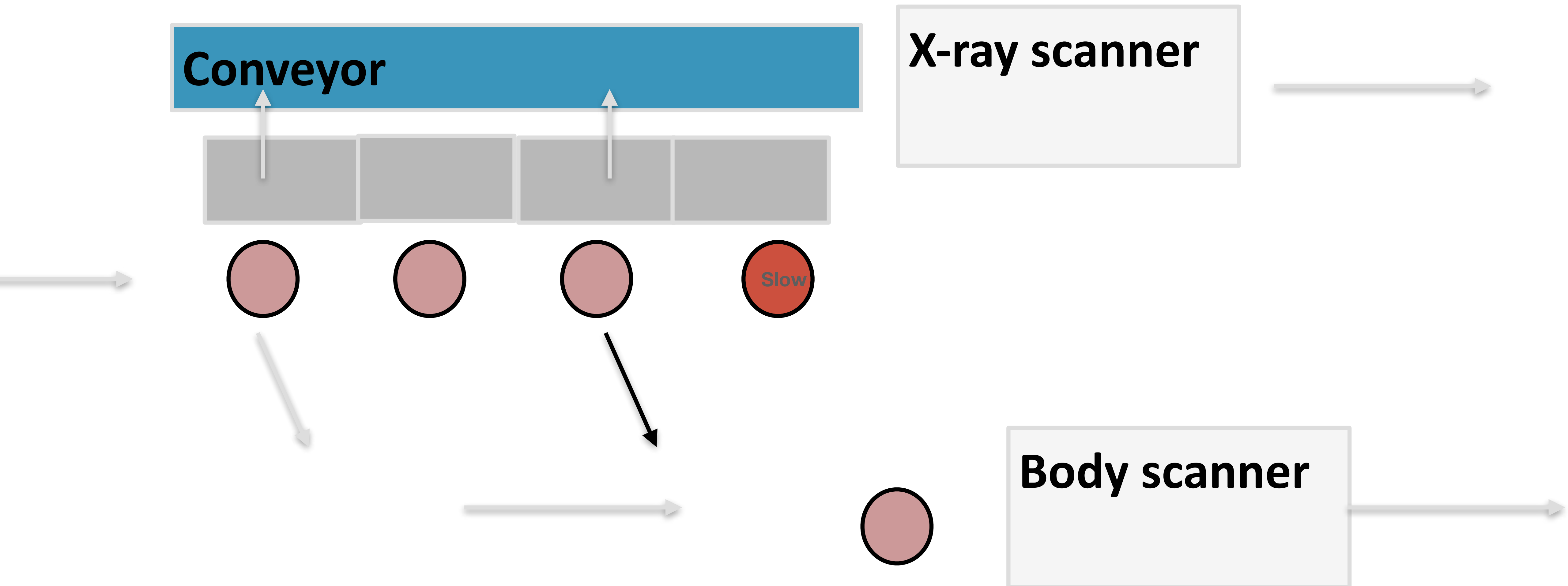
What are your suggestions for minimizing waiting time in the TSA line and significantly improve traveler experience at the airport?

Assume TSA has enough number of fast 3D scanners.

At most Airports



Good Design Solution



Enhanced security prep lanes (2020)



- Atlanta International Airport by Delta
- 30% higher throughput



- Gatwick Airport
- 30% higher throughput

MIT's 2.777, Large and Complex Systems (since 2016)



When you face a complex problem,

Where to begin?

What kind of learning do you need to solve the problem?



Principles and Disciplines

Modelling and Optimization

Heuristics and Insights



College education

Principles of Design

- Axiomatic Design
- Systems Architecture

Big national project experience

Systems Analysis

- Tenets
- Guides
- Elements

System Architecting and System Analysis (2.778)



Understand the problem

Structure the problem

Generate Concepts

Identify stake holders
Functional thinking
Solution agnostic
Find their requirements
Mind complexity

Extract/Abstract requirements
Concept of Domains
Design Axioms
Uncouple the couplings

Try a few concepts
Test them with Axioms
Reduce complexity

Test with Rigor
Story telling/Blue Team
approach

Findings at a glance (charts,
etc.)
Think one up and one down
levels
Structured Thinking

Findings at a glance (charts, etc.)
Systems Analysis

Detailed Solution Steps

MS1

MS2

MS3

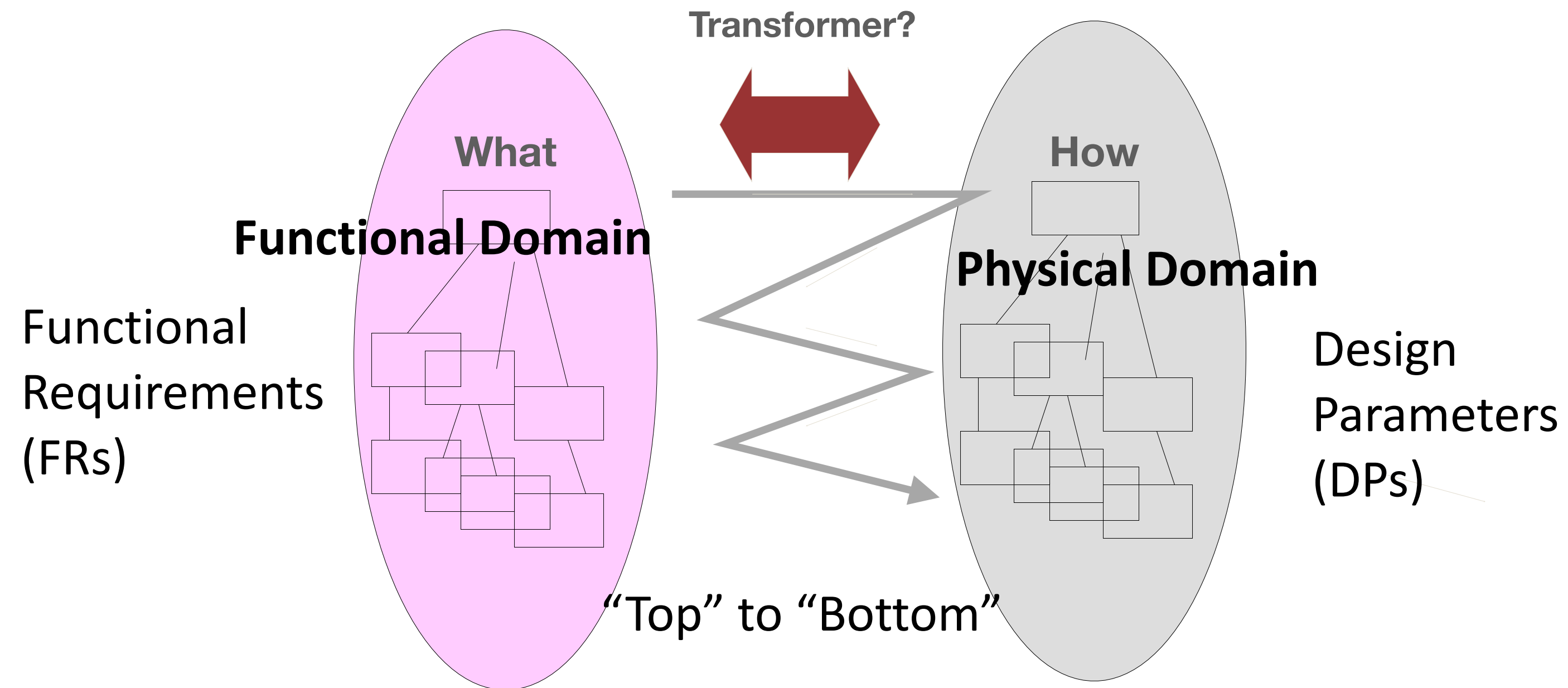
MS4

MS5

Lectures

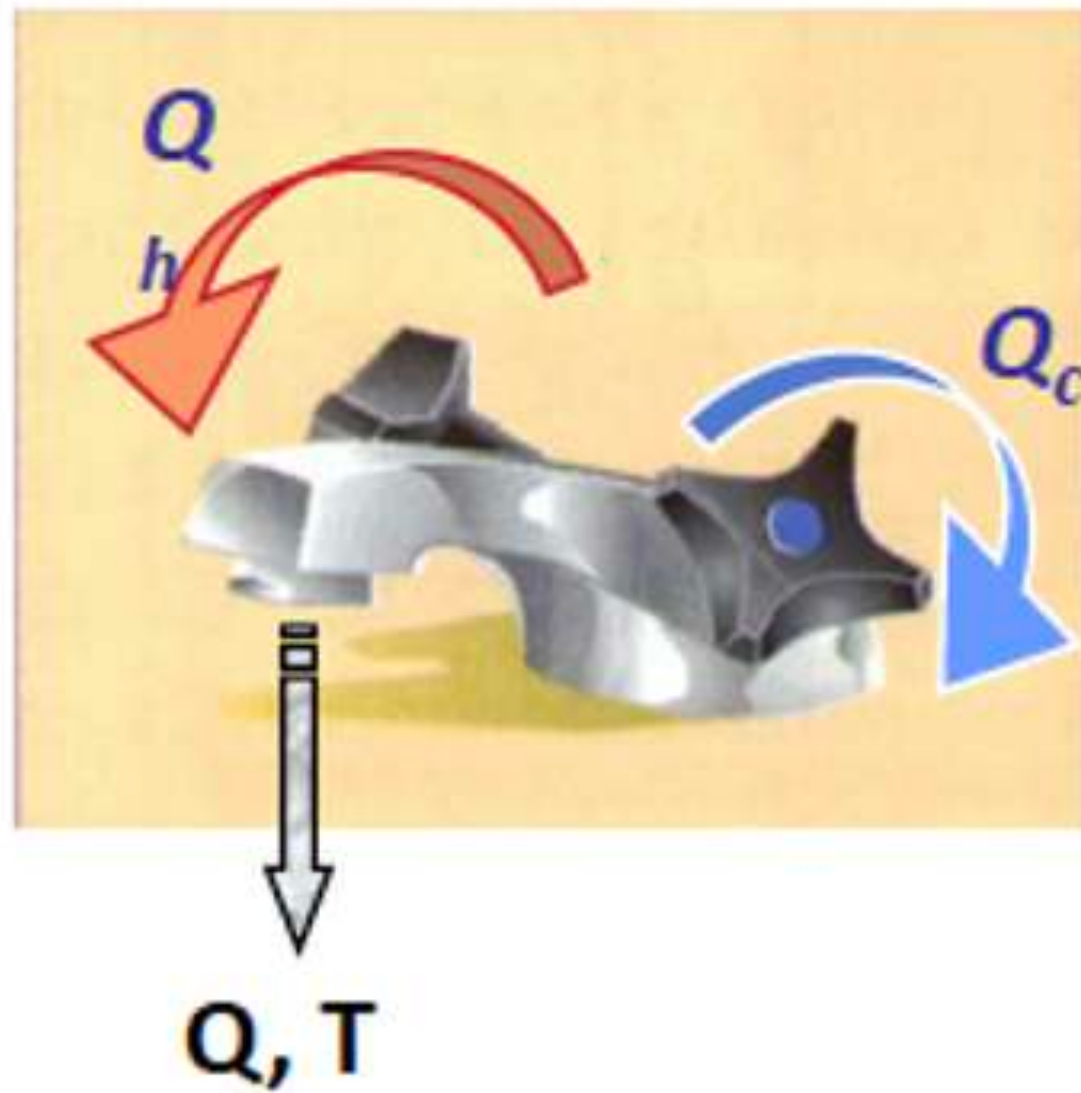
Projects

Aximmatic Design



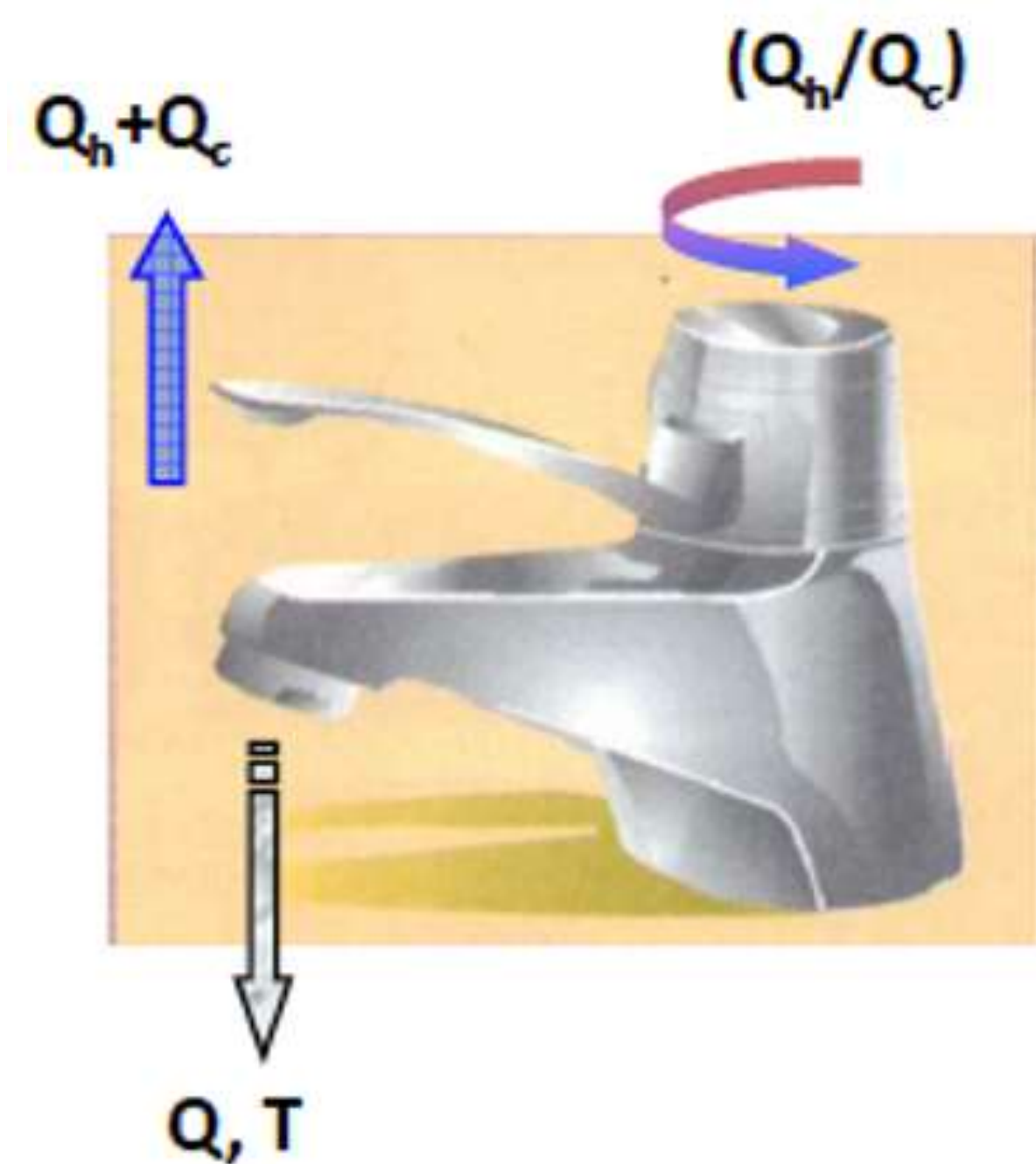
Observation #1: Functional Couplings are bad.

Fully coupled design



FR \ DP	Hot water flow (Q_{hot})	Cold water flow (Q_{cold})
Control the flow (Q)		
Control temperature (T)		

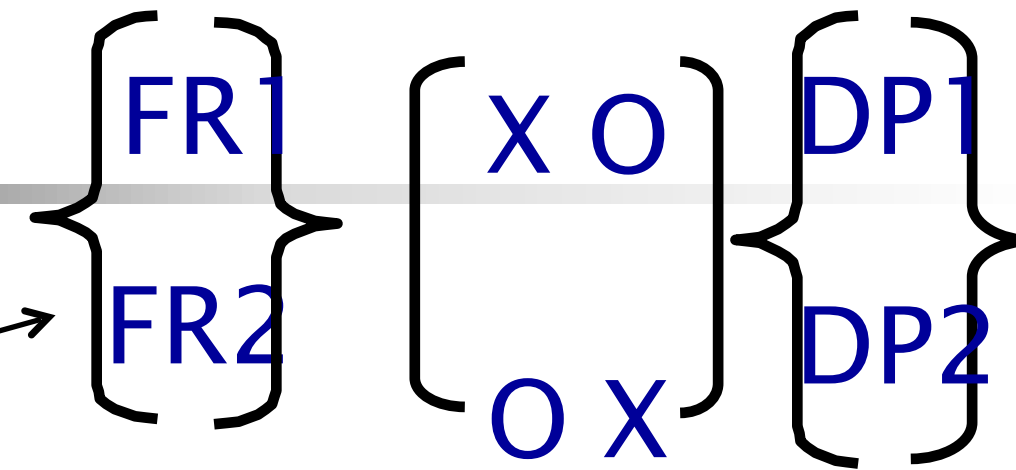
Uncoupled Design



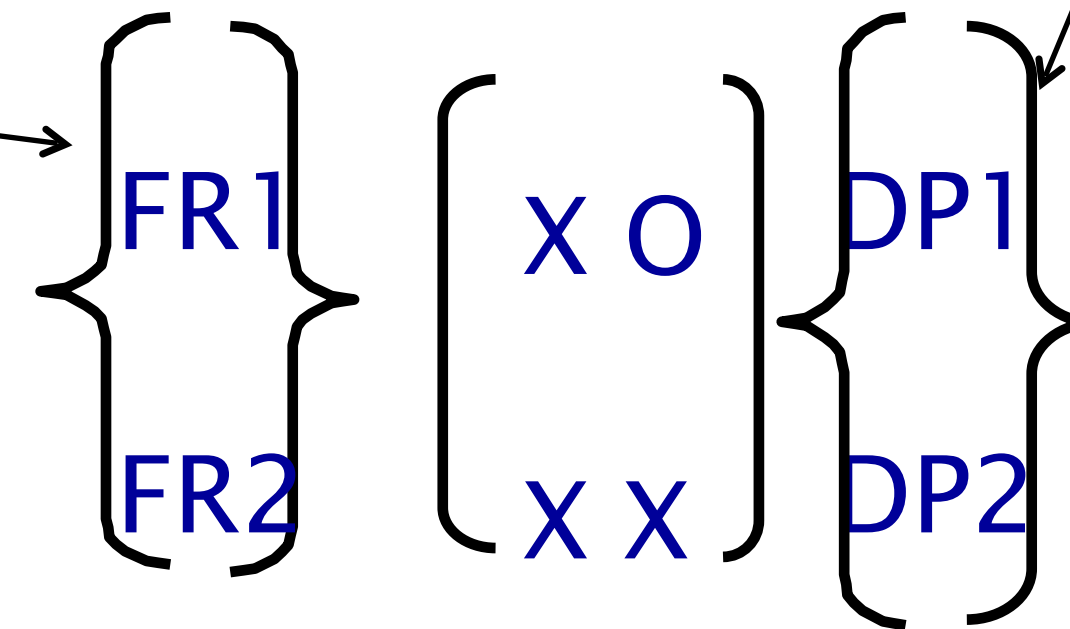
	Flow control V-level, $Q_{hot} + Q_{cold}$ ($Q_{hot}/Q_{cold} = C$)	Temp control by rotation, Q_{hot}/Q_{cold} ($Q_{hot} + Q_{cold} = C$)
Control the flow (Q)		
Control temperature (T)		

Functional Couplings

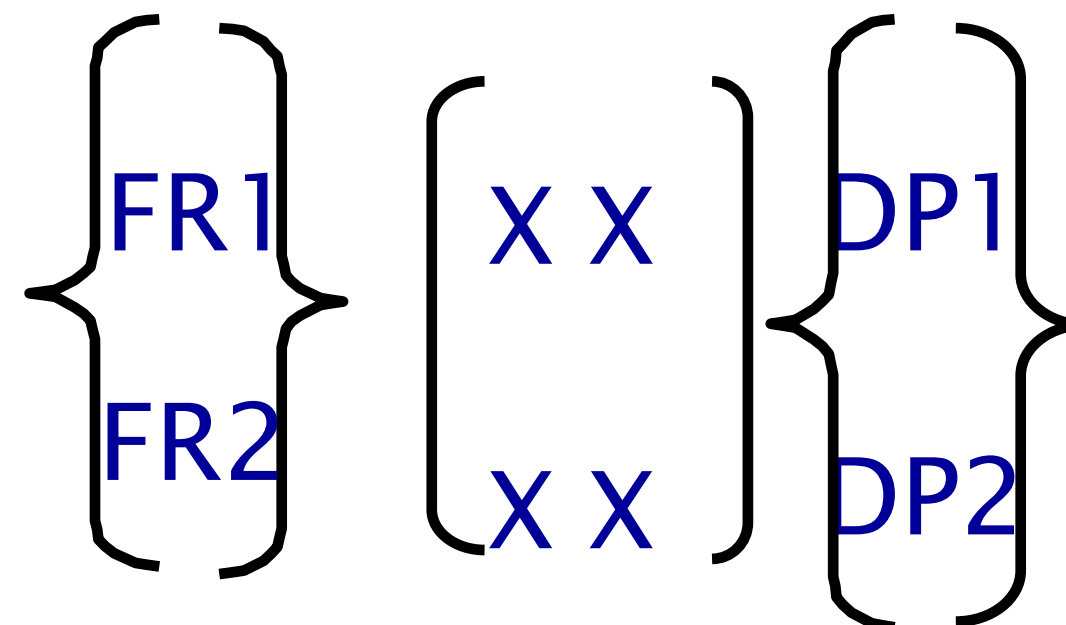
■ Uncoupled



■ Decoupled



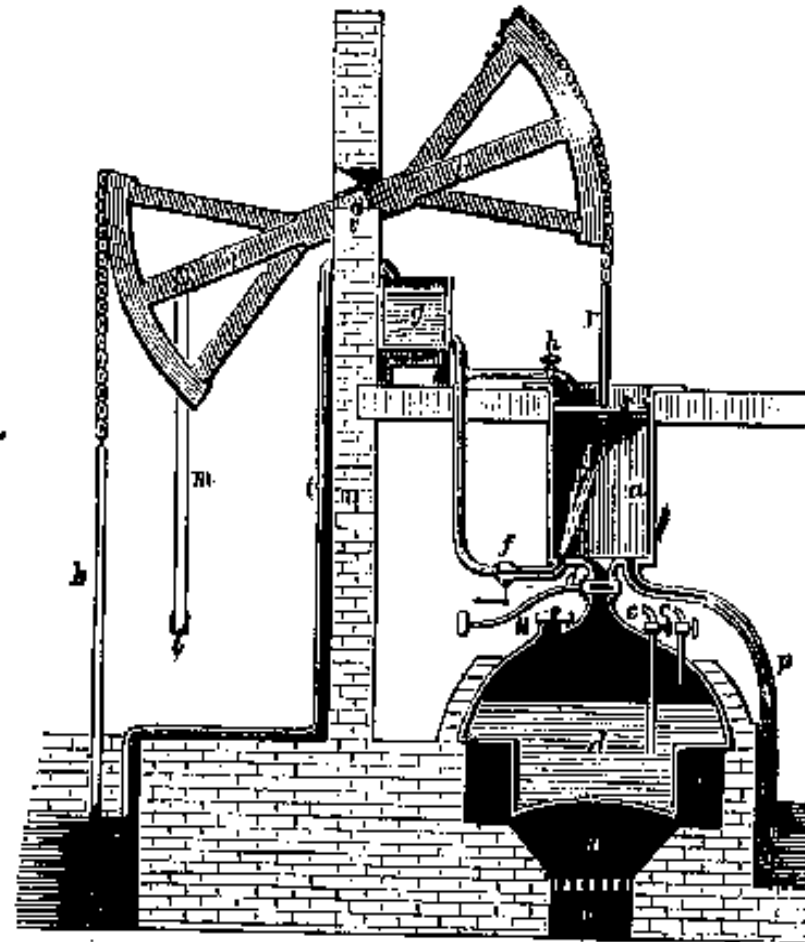
■ Coupled



Bad Design

Observation #2: If you see functional coupling early stage of your design, big big savings of resources.

Axiomatic View of Watt's Steam Engine Design



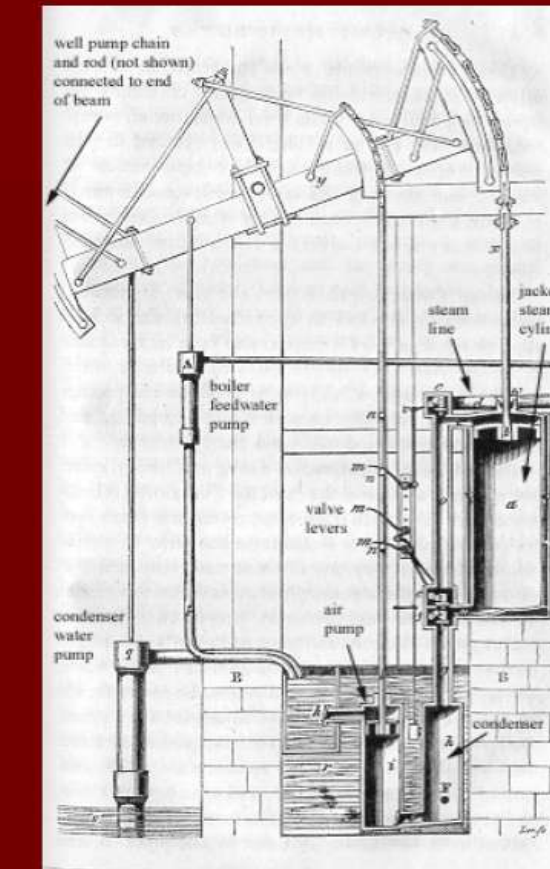
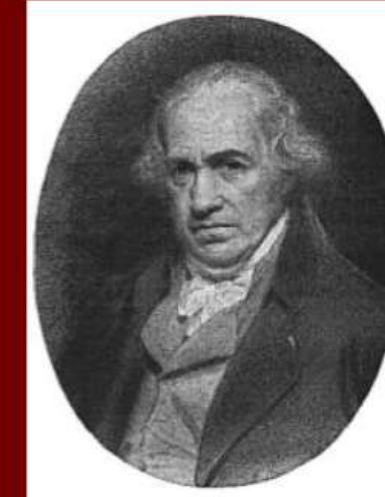
Newcomen's Engine, A. D. 1705.

Watt's Single-Acting Steam Engine

Greatly improved efficiency of Newcomen's engine (1769)

Jacketed steam cylinder and separate condenser

"Catch-all" patents prevented further progress by other engineers



Watt's Engine, A. D. 1769.

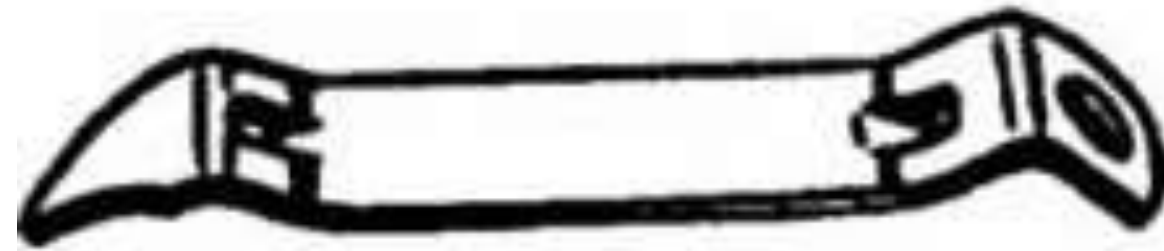
- First Law of Thermodynamics, 1850
- Second Law of Thermodynamics, 1824-1860, Carnot, Clausius

$$\eta = \frac{\Delta W}{\Delta Q_H} = 1 - \frac{T_C}{T_H}$$

Carnot Theorem

Coupled design or not?

Can and bottle opener



Observation #3: Physical coupling is good in most cases. Functional coupling is bad.

Case: Solar Car Race 1987



Sunraycer (1987)



Gossamer Albatross

1979



Pathfinder
Helios

1982

Future versions of the unmanned, solar-powered Helios will fly above 50,000 feet for six months at a time, serving many purposes, such as telecommunications and weather watching.



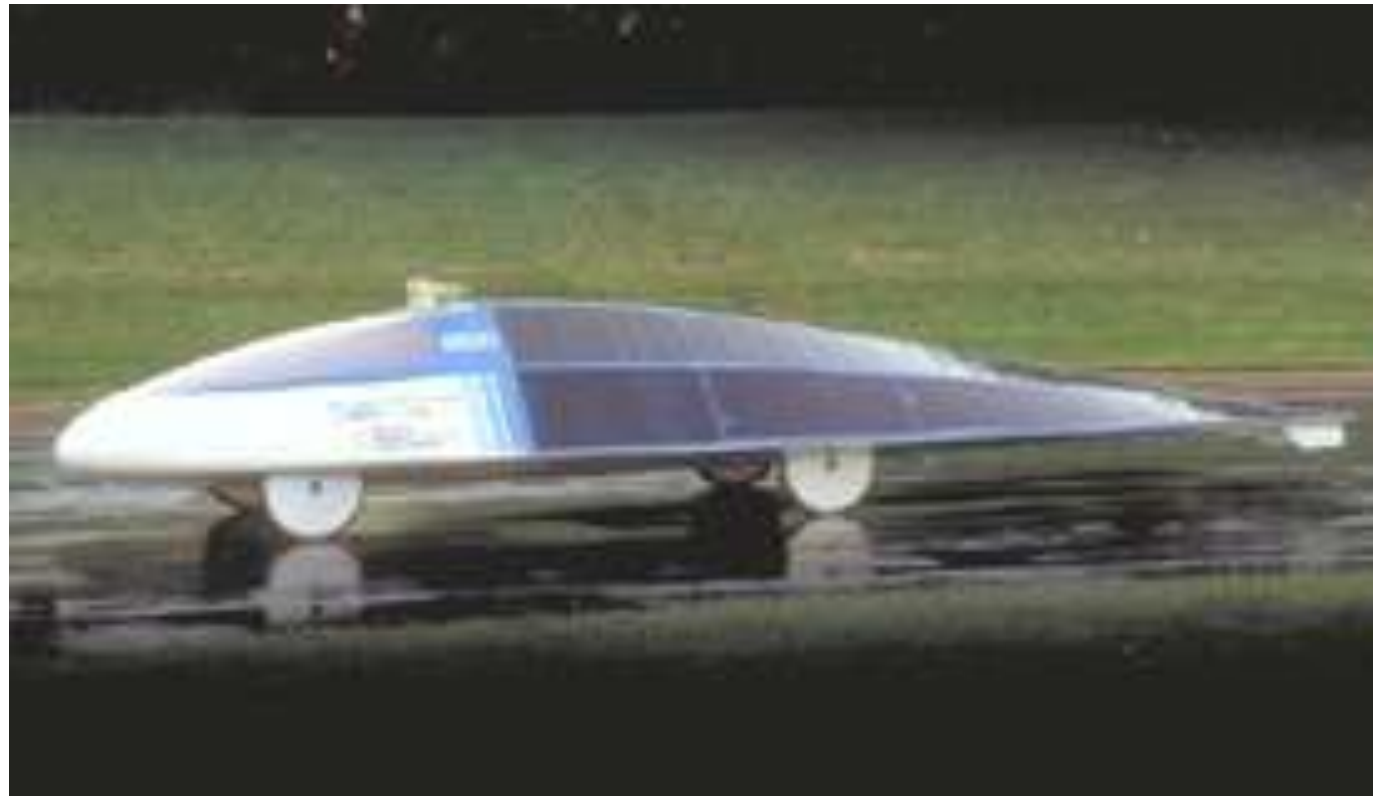
Paul MacCready, Aerovironment
(1925-2007)



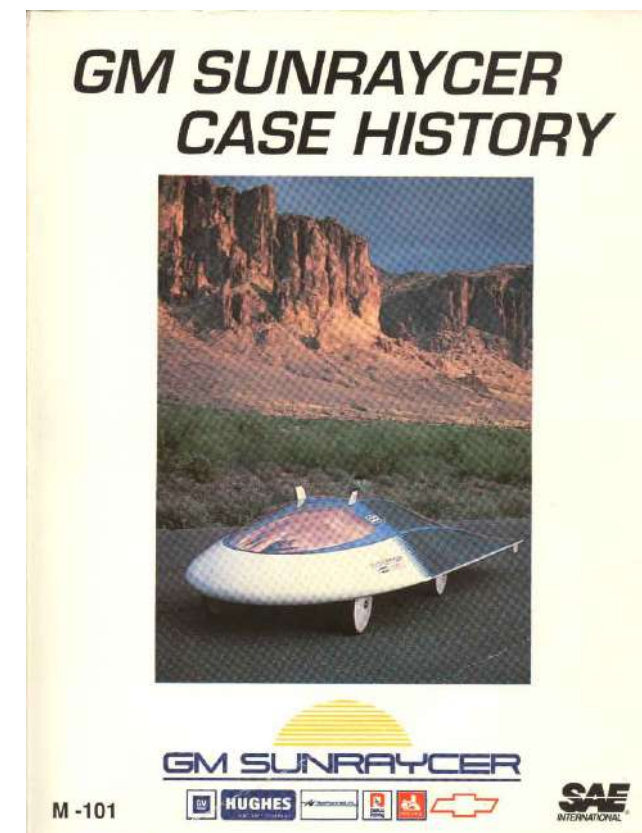
Mars Helicopter, Ingenuity

Sunraycer

Paul MacCready, 1987, Aerovironment Co.

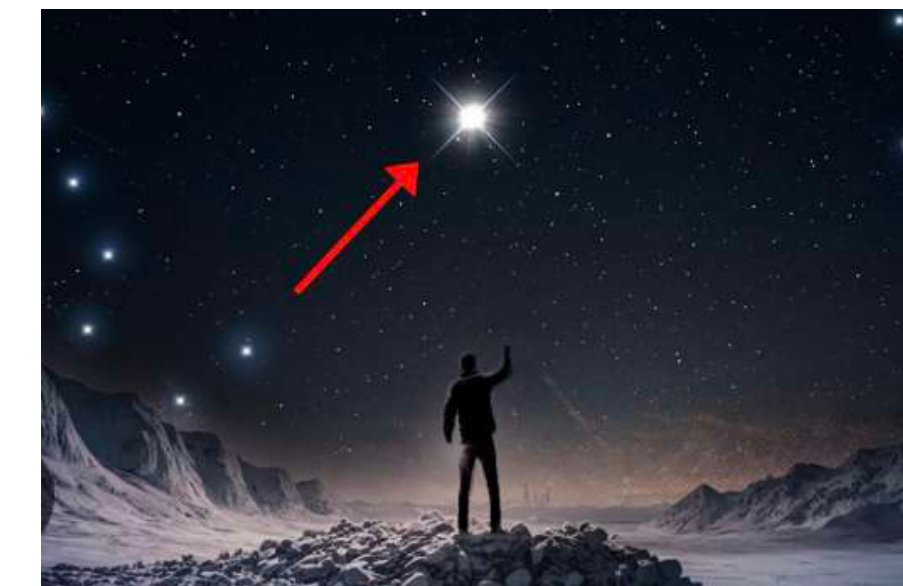


GM Impact



4 Top Winning Strategies:

Light weight
Efficiency
Reliability
Simplicity



Observation #3: Design is Top-Down. North star of design goals

Design Principles of Axiomatic Design Thinking

Observation # 1: Functional Couplings are bad.

Observation # 2: Uncoupling the function coupling often results in drastic improvements under time pressure. A way to handle system complexity.

Observation # 3: Physical coupling is good in most cases unless functionally coupled..

Observation # 4: Design is Top-Down. Find **north star** of design goals.

Case: Social system design — Health Care



Partners Healthcare (now Mass General Brigham), ~75,000 staffs

Can we apply the Toyota Manufacturing System to Hospitals?

ER crowdedness (Newton Wellesley Hospital)

MAXWELL BLUM EMERGENCY PAVILION

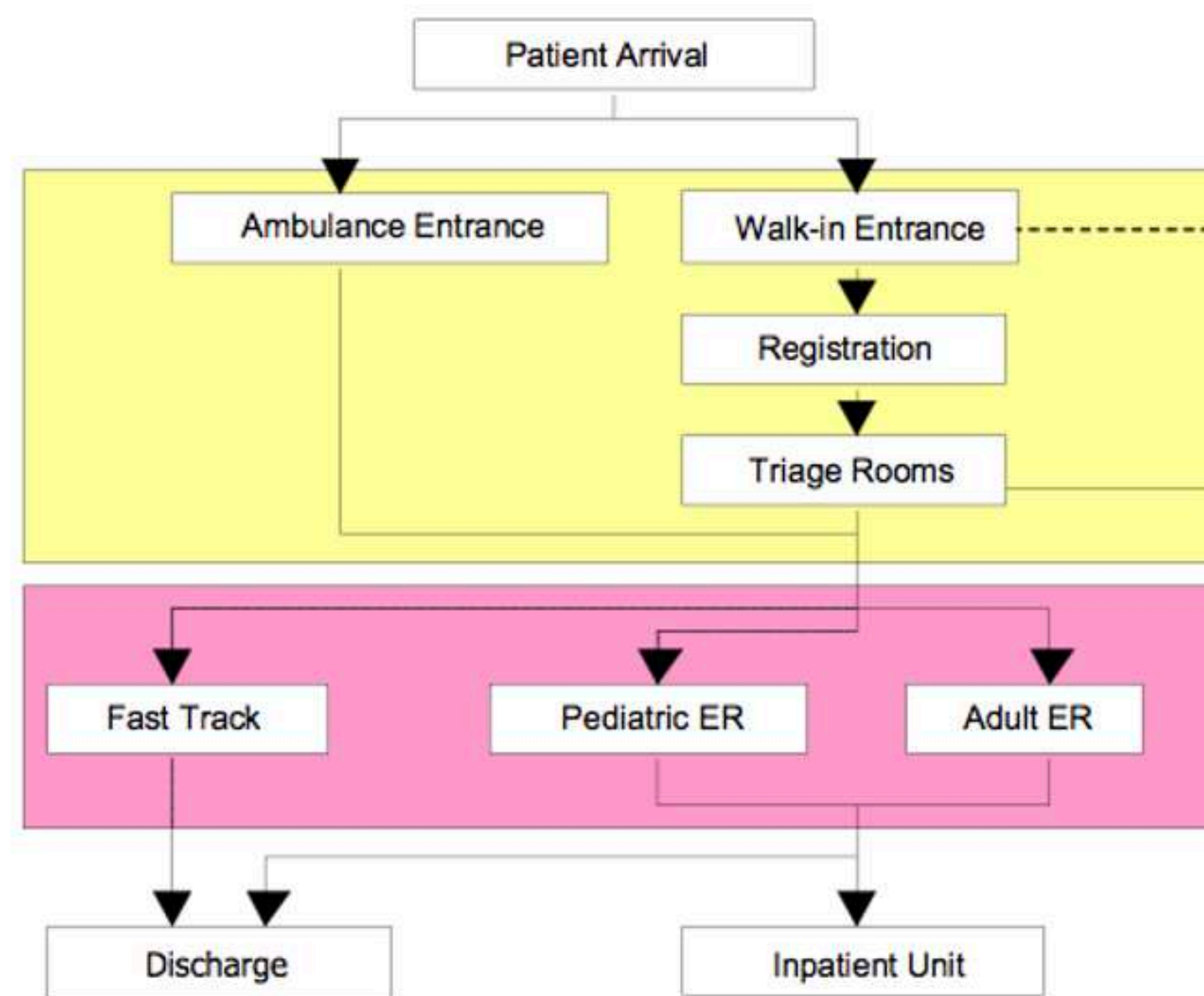
- Background
 - Over the past six years, ED visits at NWH have soared from about 30,000 to just under 50,000
 - Long waits into a treatment room: avg 55 min on weekdays, avg 90 min on weekend; national overall average is 56 min (as of 2004)
- 45 minutes weekdays and 55 minutes on weekends (on NWH web page)



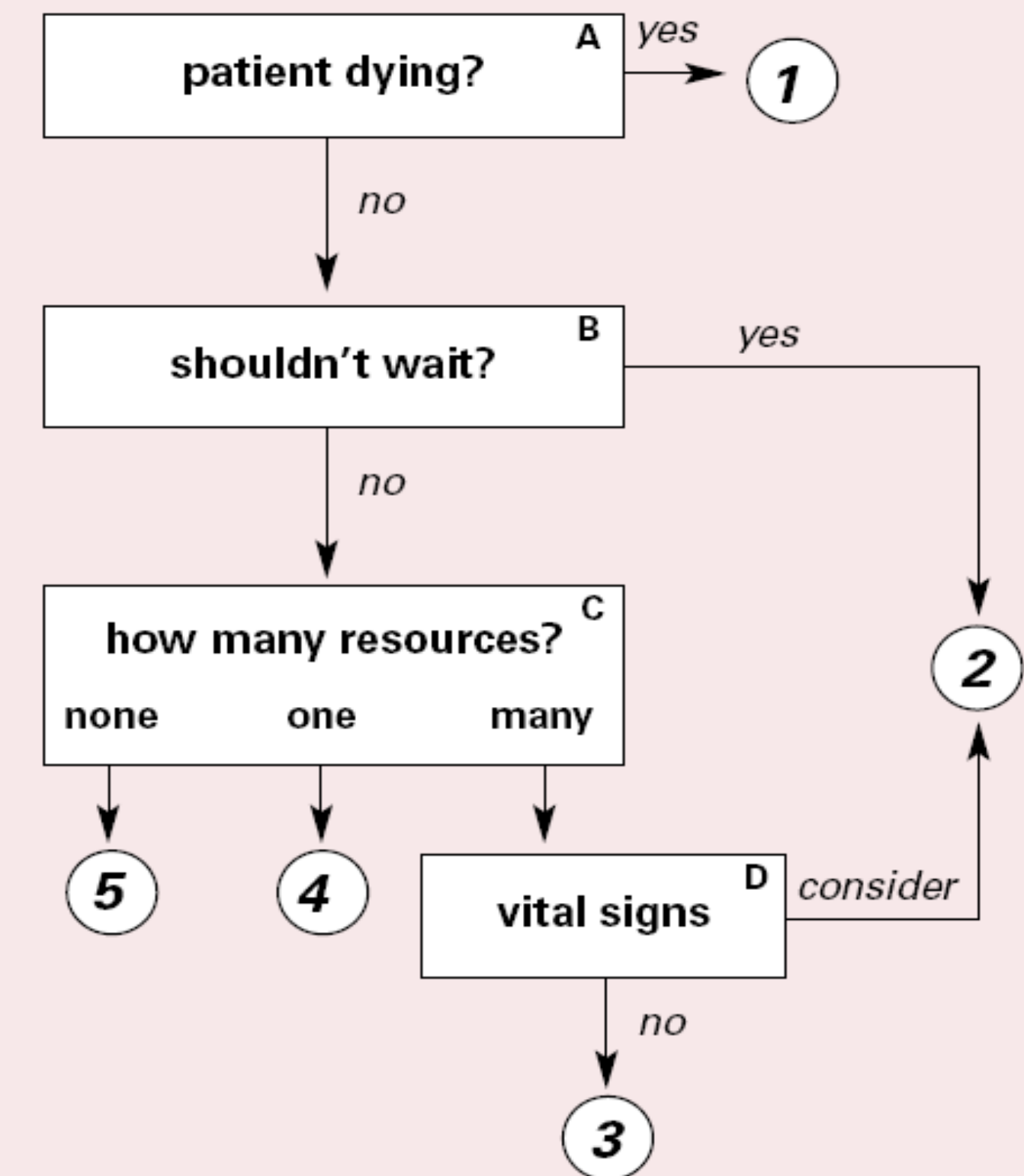
	Old ED	Maxwell Blum EP
Target volume	28,000/yr	55,000/yr
Footprint	~12,000 sq. ft	35,000 sq. ft
Private rooms	19	36
Pediatric section		8 beds
Adult section		24 beds
Fast-track	3 beds	4 beds
Fast-track op hr	? Hrs	8 hrs(wk) /10 hrs(wknd)
Employees	X	X + 47
Triage room	Y	Y + 1
Test equipment	?	Two X-ray rms, a CT scanner, ultrasound scanning

Fast Track Problem

- FT serves the purpose of increasing ED Capacity and if it is not used to its maximum then resources are being wasted.
- FT is used to decrease the waiting time of low acuity patients, yet despite the existence of FT the ESI 4 and 5 TTB remains the highest.

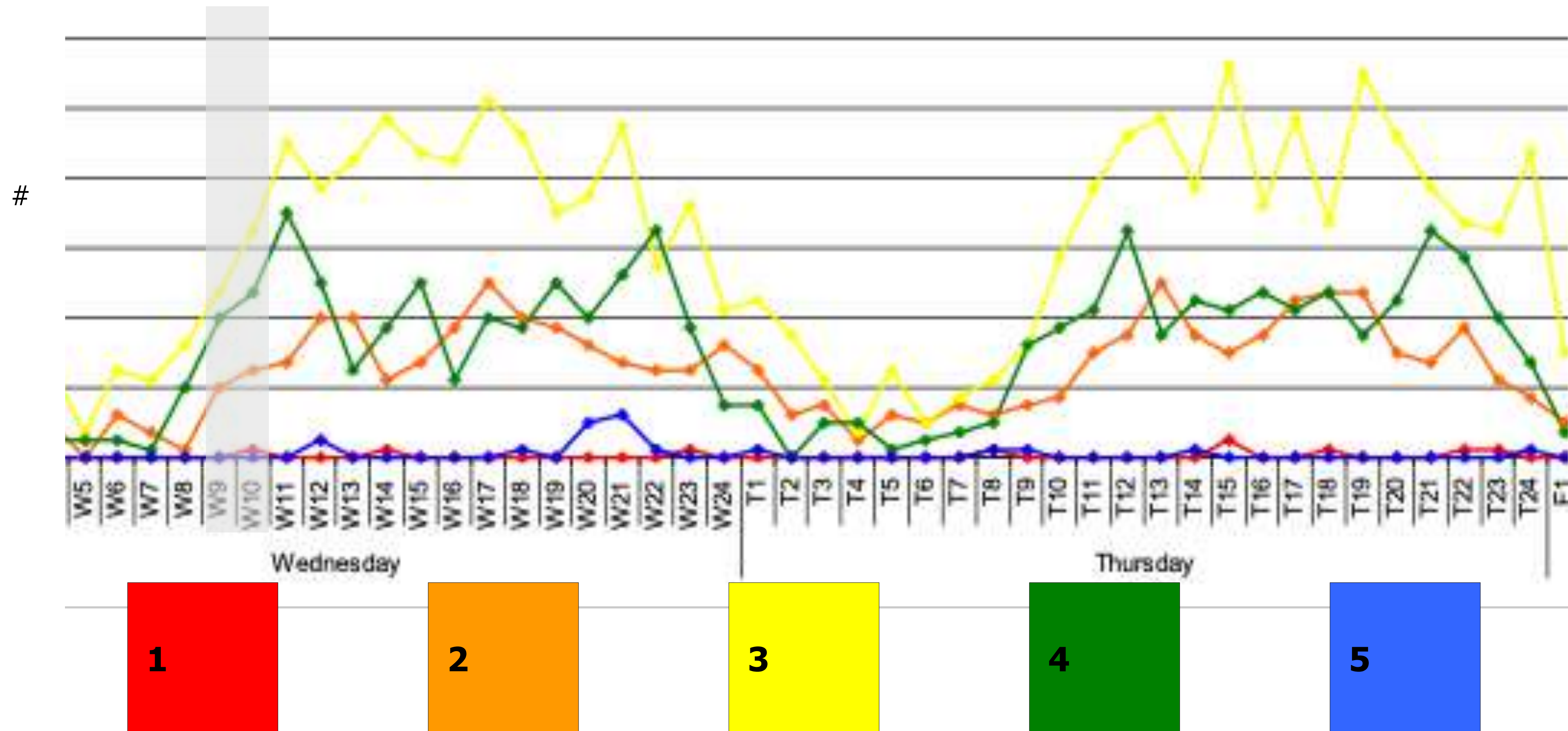


Triage of ER



©ESI Triage Research Team, 2004. Reproduced with permission.

The Emergency Department Patient Flow

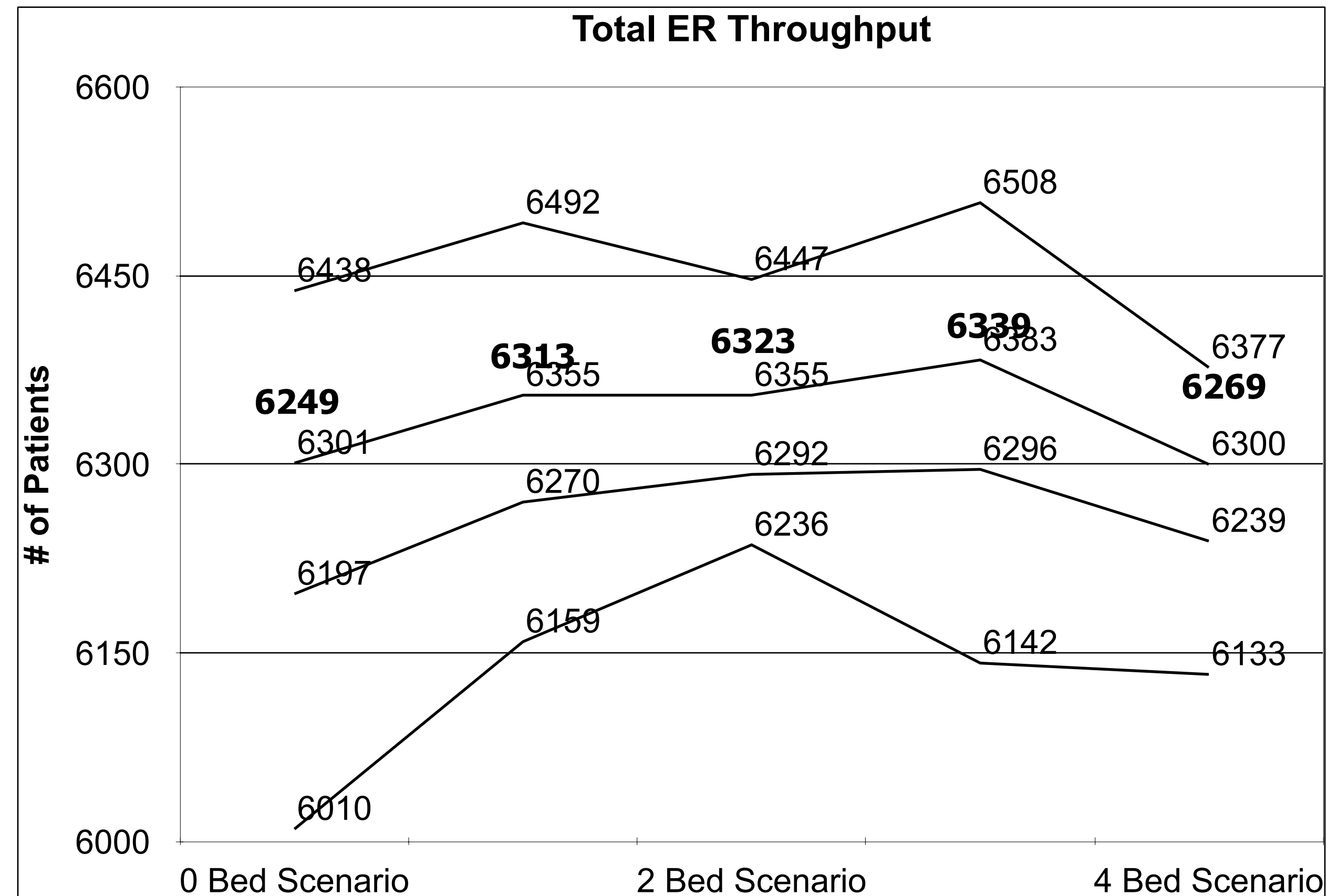


- **Emergency Severity Index (ESI)**—a five-level emergency department triage algorithm that provides clinically relevant stratification of patients into five groups from 1 (most urgent) to 5 (least urgent) on the basis of acuity and resource needs.

First Approach

- Optimization
 - ED³ (Emergency Department Design Decomposition)
 - Discrete-Event Simulation model using ARENA software package
 - Number of FT beds (# out of 28 beds)
 - Vary ESI allowed to Fast Track
 - 12,000 ED patient data over 8 months

No meaningful Improvement with 3 weeks left for final report

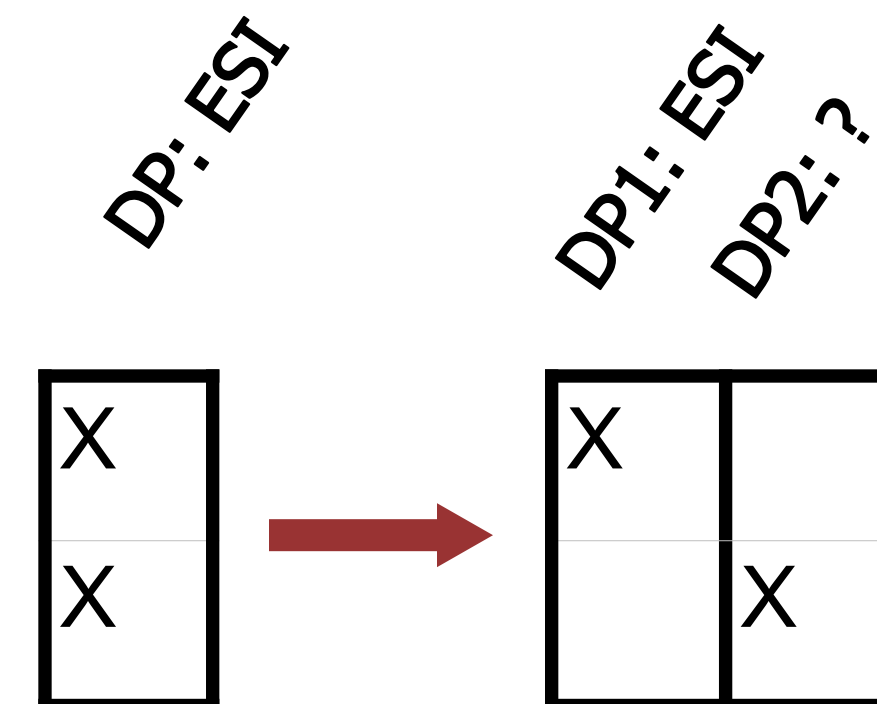


Possible trend, though it is not obvious, we need to break down the throughput for better understanding.

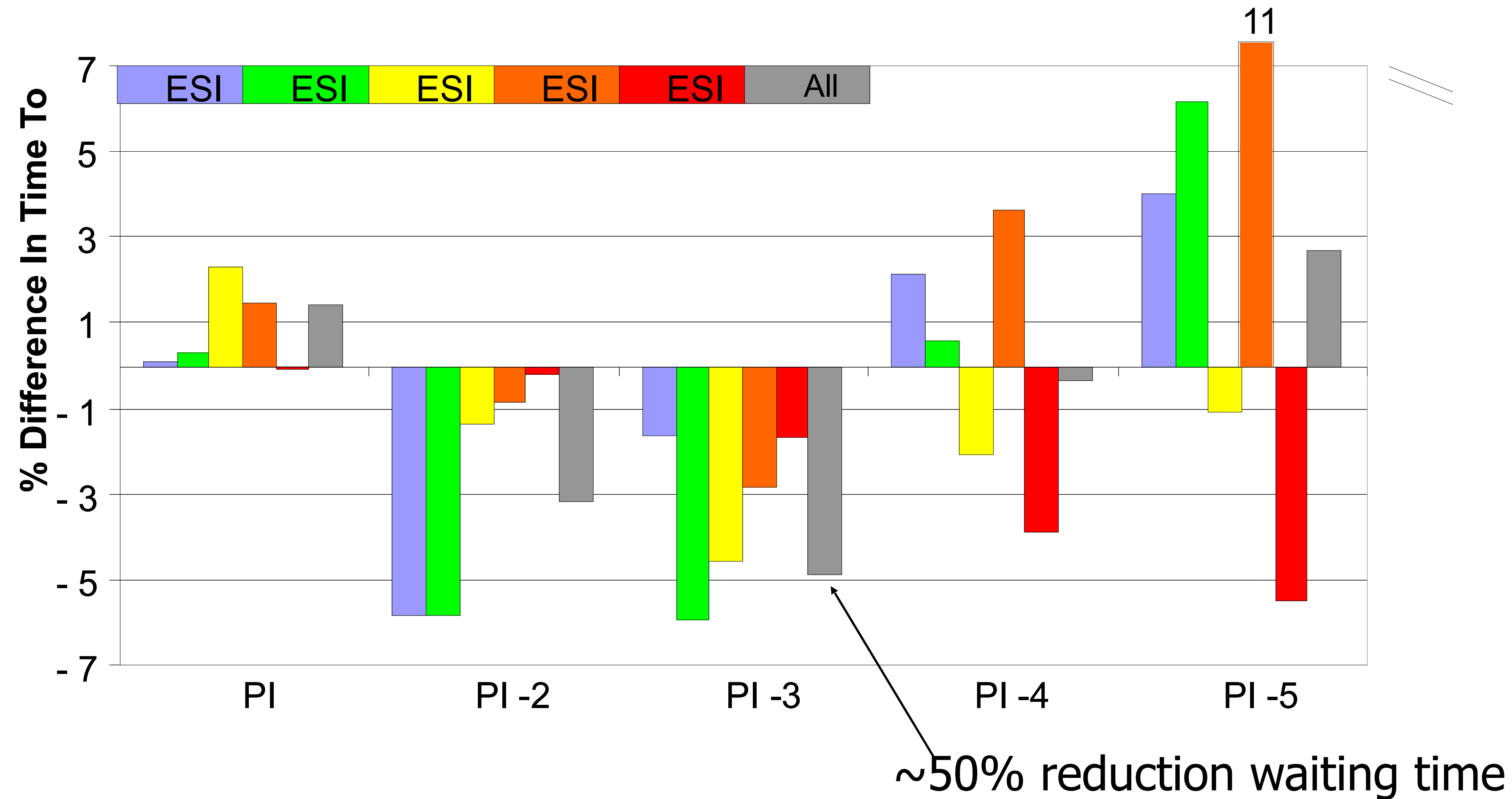
- A Triage Nurse assigns an ESI based on how urgently the patient needs care, and how many resources they will require. The amount of time a patient will spend in ED is not taken into account.
- In practice, patient treatment time is estimated based on their ESI, meaning that DP1 is used to satisfy both FR1 and FR 2, creating a coupling.

FR1: Prioritize Patients Based on Treatment Urgency

FR2: Sort Patients to Facilitate Flow



Percent Difference in Time to Bed from 28 Bed Scenario



Observation #2: Uncoupling the functional coupling results in drastic improvements.

J. Peck, S-G. Kim, "Improving Patient Flow through Axiomatic Design of Hospital Emergency Departments," Journal of Manufacturing Science and Technology, Vol.2, Issue 4, P. 255-260, 2010, ACEP Research Forum, American College of Emergency Physicians Scientific Assembly, Chicago, IL, 2008

© Sang-Gook Kim, MIT



Reading Material on Canvas

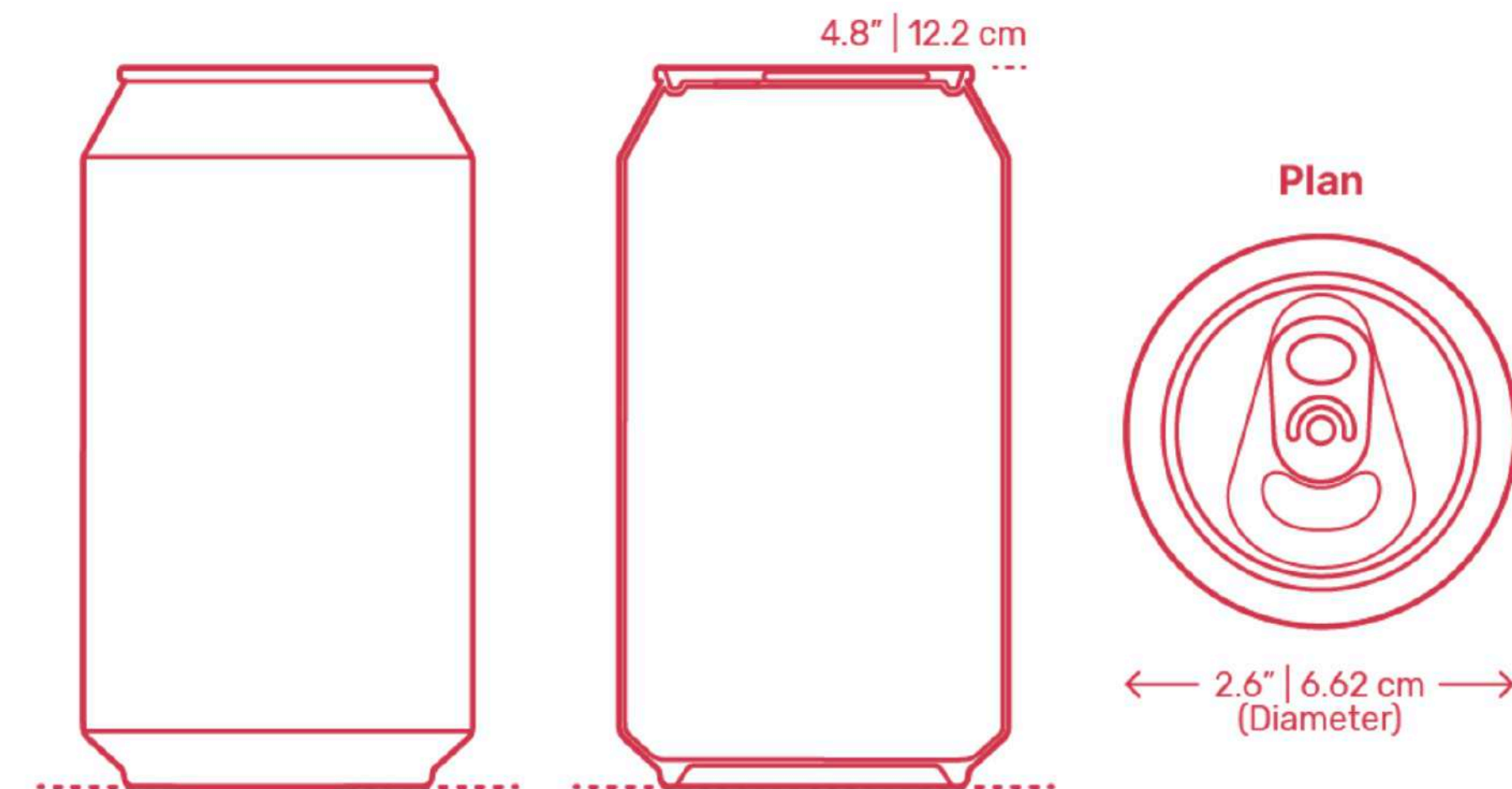


Finding FRs

- As a designer, you are completely free to define your problem in any way.
- By definition, each FR is independent of every other FR at the time FRs are established.
- All FRs are equally important.

Example: Beverage Can Design

- Consider an aluminum beverage can that contains carbonated drinks.
- How many functional requirements must a can satisfy?
- What are the design parameters (DPs)? How many DPs are there?
- How many physical parts does it have?



Functional Requirements (FRs)

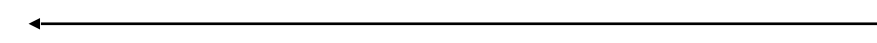
FRs of a aluminum beverage belonging to the main body piece

- Contain carbonated fluid
- with no leakage (sealing)
- Opening
- ...

- Cost of material ??

FRs of a aluminum beverage belonging to the main body piece

Contain carbonated fluid



Al can

Retain structural rigidity

- axial, radial

Opening easily

Compact packing

Resistance to rust

Printable

Flow rate, Drinkability

Safety for drinking

Cleanness, hygiene

Tight sealing, gas Leaking rate

Light weight

Cost of material ??

(curvature, thickness)

(pop tab shape)

(horizontally, vertically)

(inside, outside)

Constraint or FR ?

FRs of an aluminum beverage can

Containing carbonated fluid

Retain structural rigidity

- Axial stiffness
- Radial stiffness
- ..

Safe drinking

- Opening effort
- Flow rate
- Sharpness
- Internal coating
- ..

Decoration

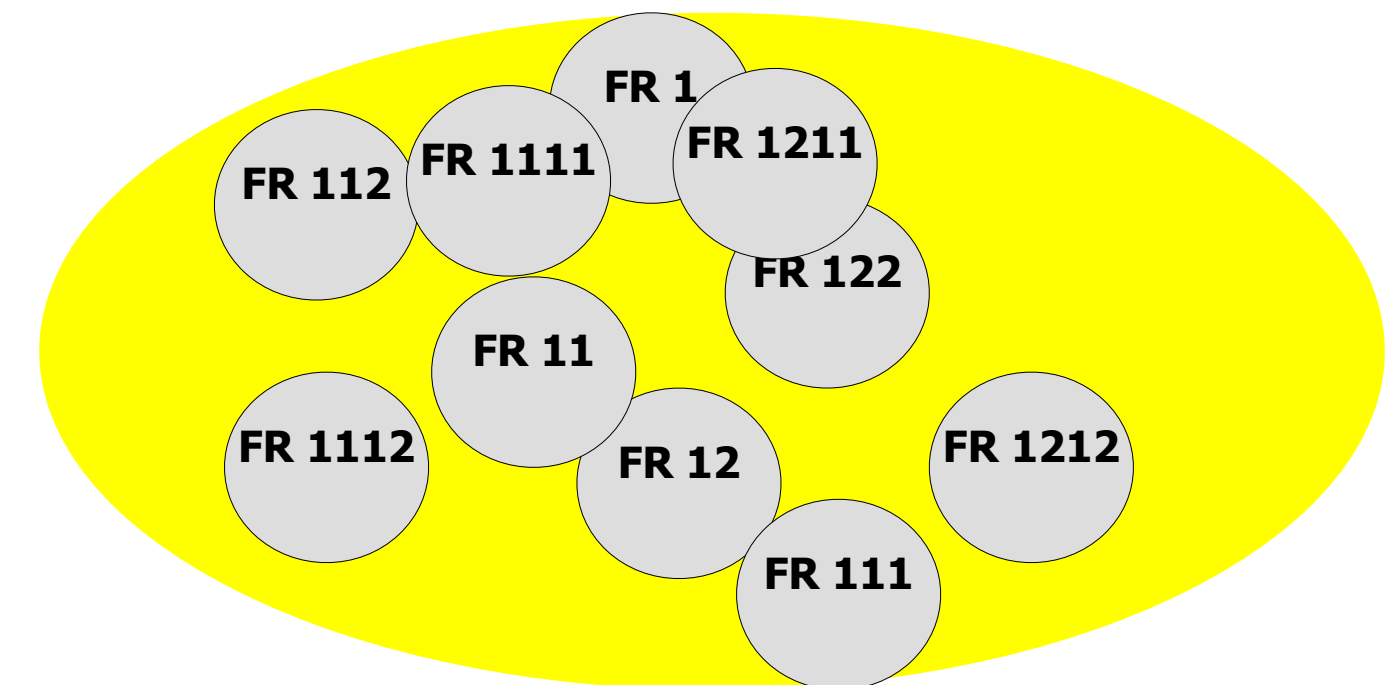
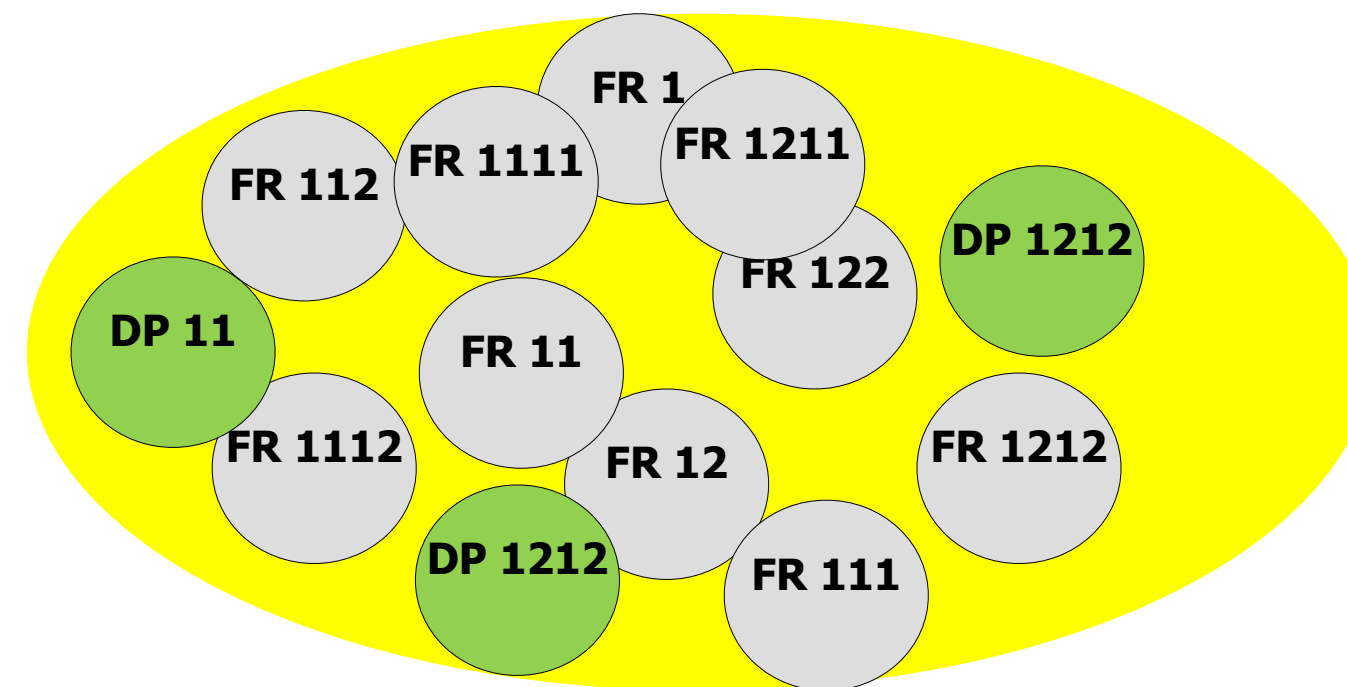
- Printable
- Slender shape
- ..

Distribution

- Stackable
- Packing density
- ..

Understanding the problem

- Recognize two domains of design.
- Separate “What’s (FRs)” from “How’s (DPs)”.



FRs: functional requirements
DPs: Design parameters

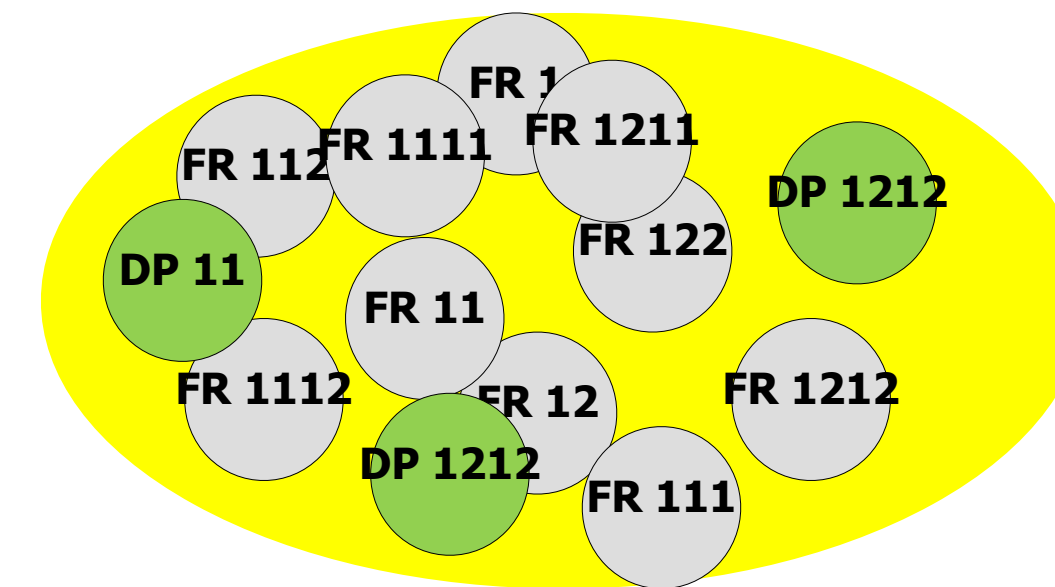
Can AI think in Axiomatic Thinking

DO's

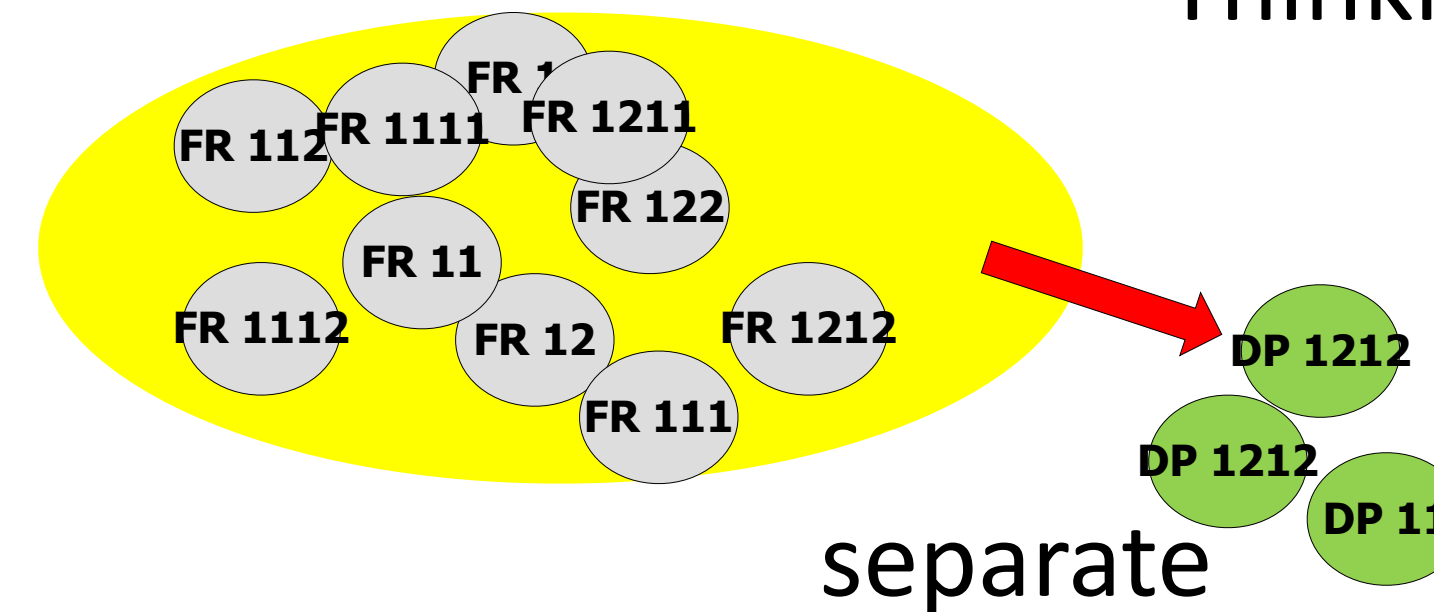
- Figure out users and stake holders' needs, and constraints.
- Focus on "what's"
- Put aside "how's"

Don'ts

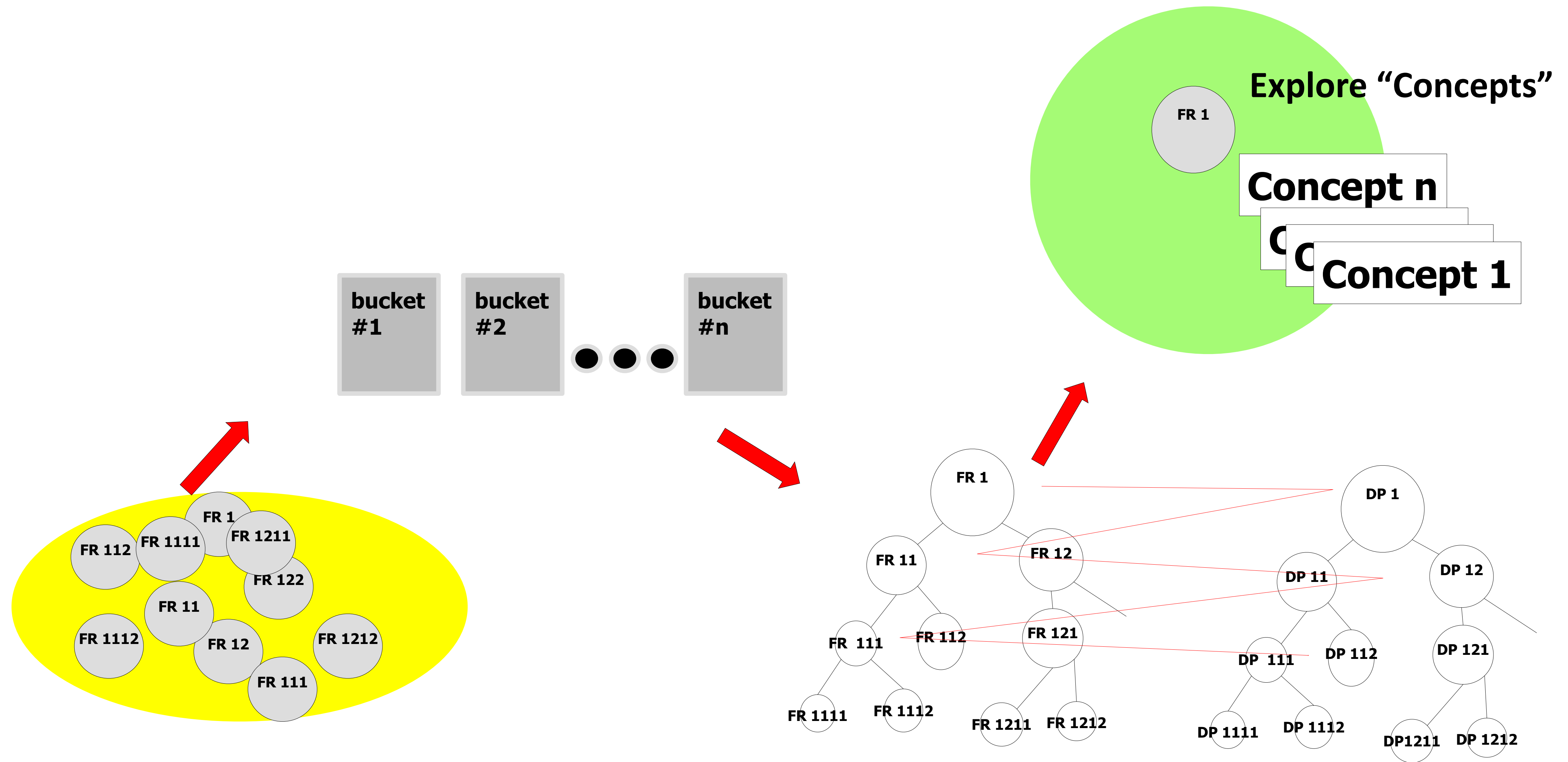
- Do not mix "what's" and "how's"
- Not solution neutral



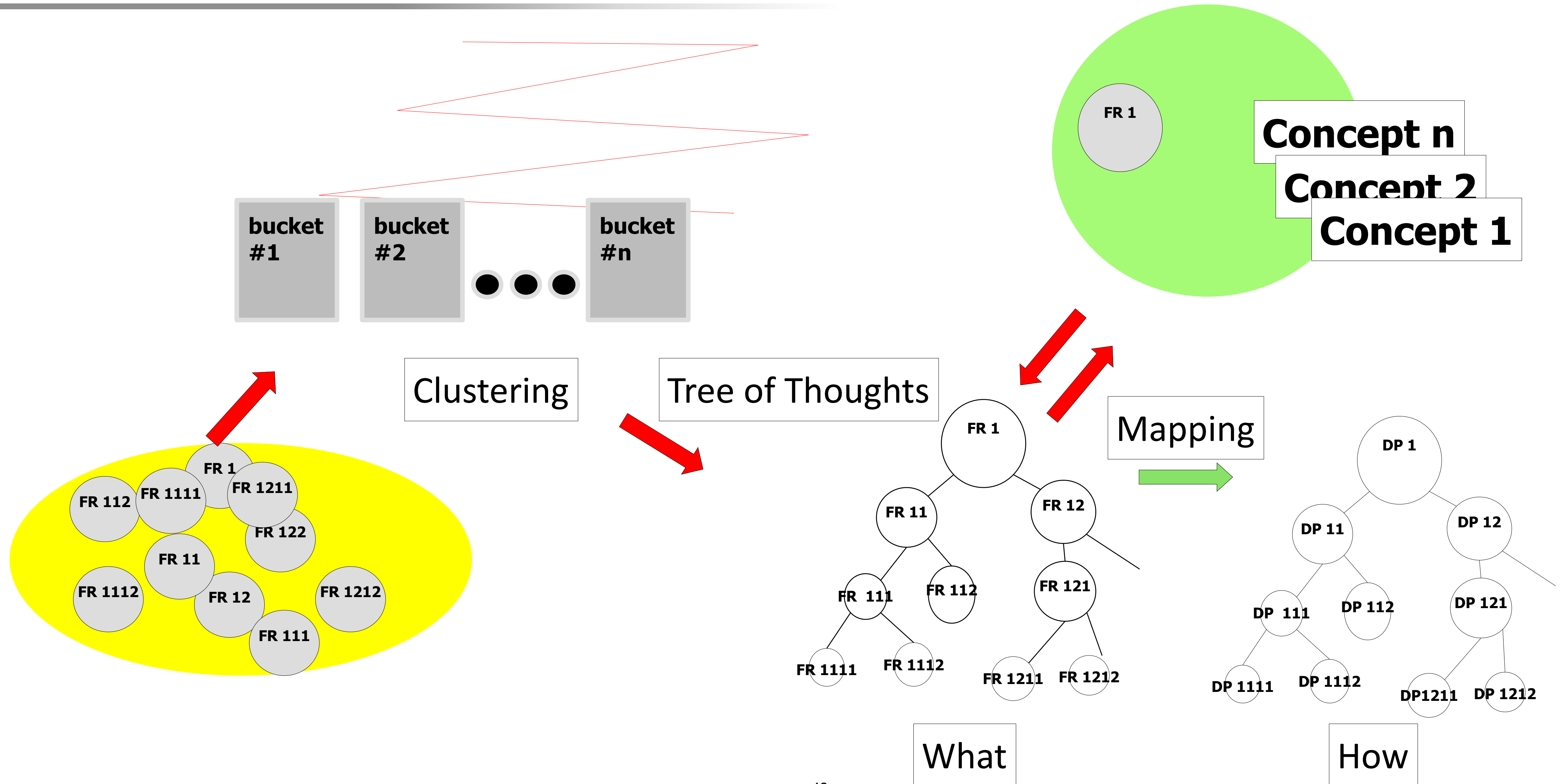
Functional Thinking



Solution Agnostic



Can AI abstract findings and generate concept?



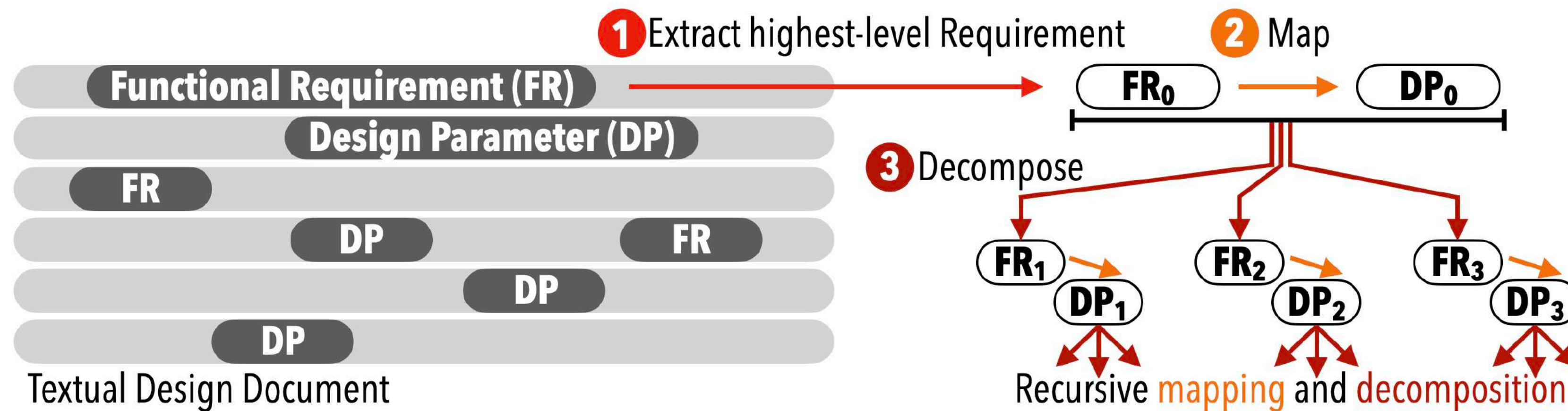


Can I train AI think in Axiomatic way?

Thinking Design Machine

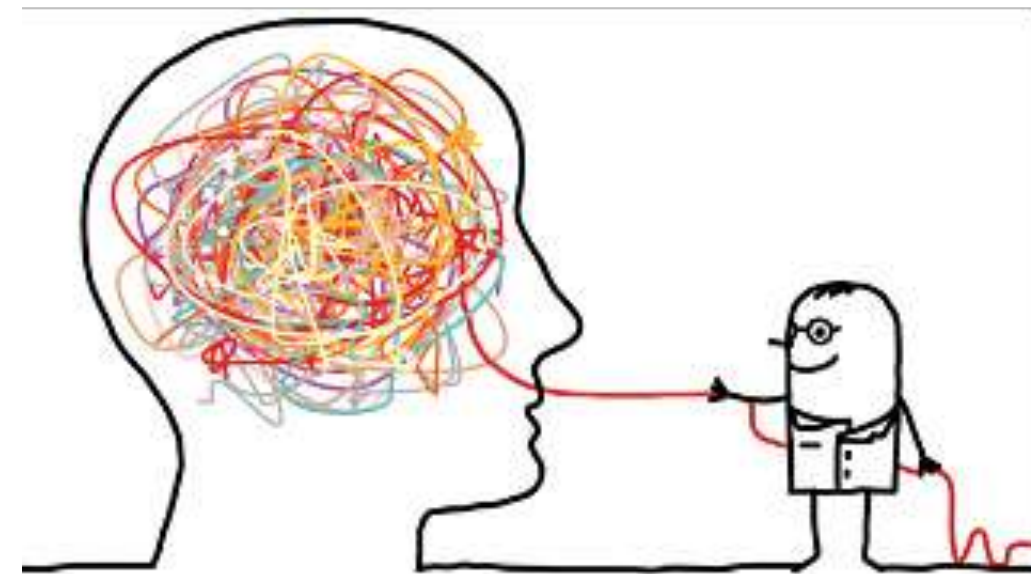
Extracting functional decisions in design and manufacturing

- Process for extracting functional structure from text document

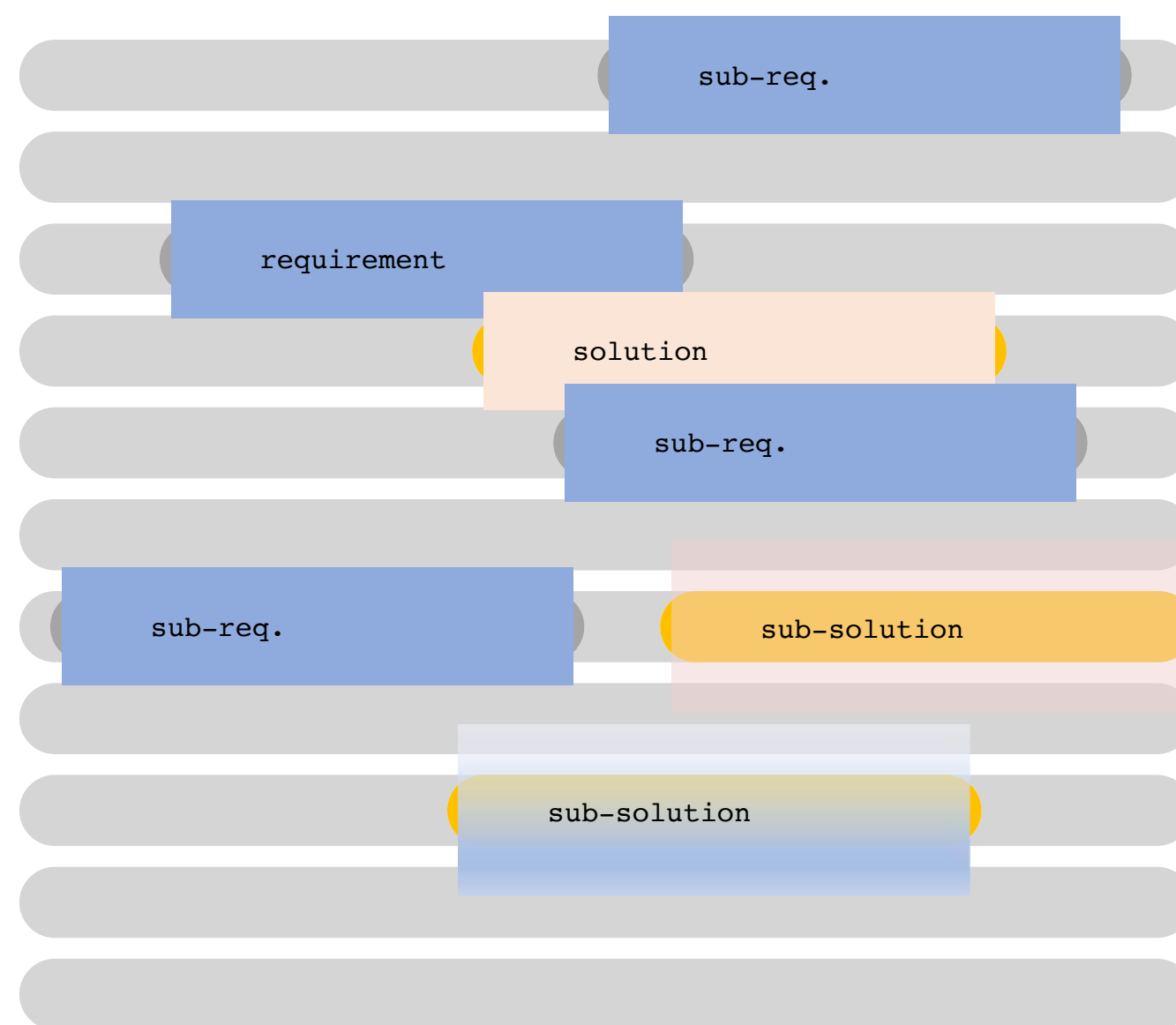


"Tree of Thoughts" reading from design documents

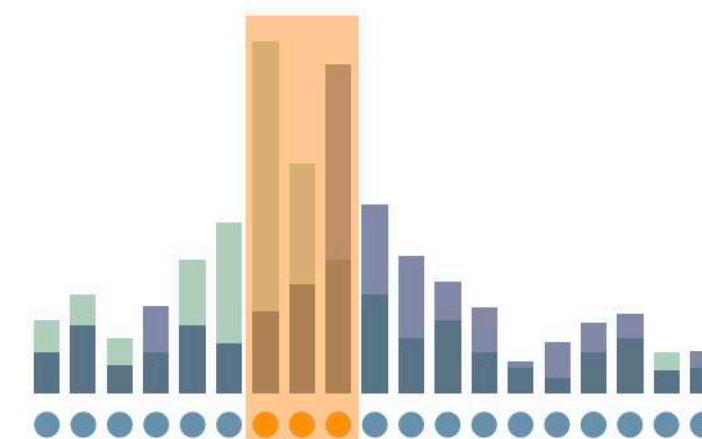
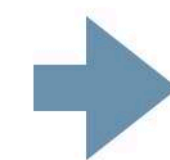
Recursive 'question and answer' to extract the design decisions, FRs and DPs.



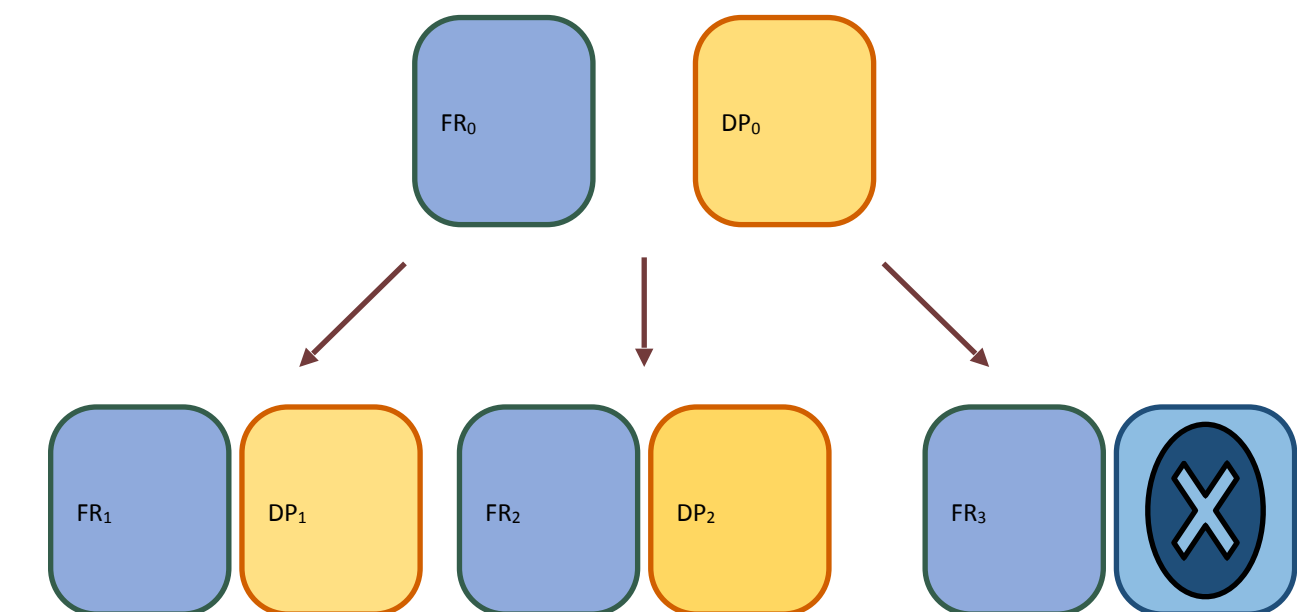
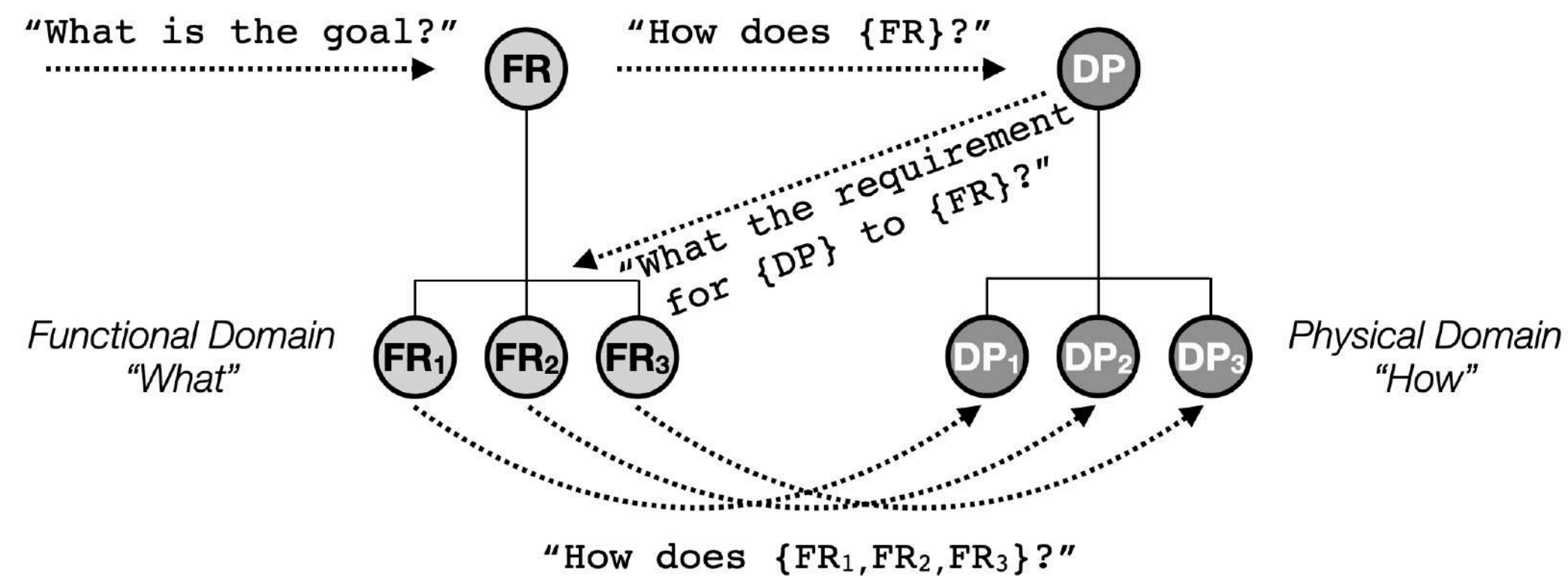
Documented design



Start
Position
End
Position



Sequence of vectors
(Length = # words in document)

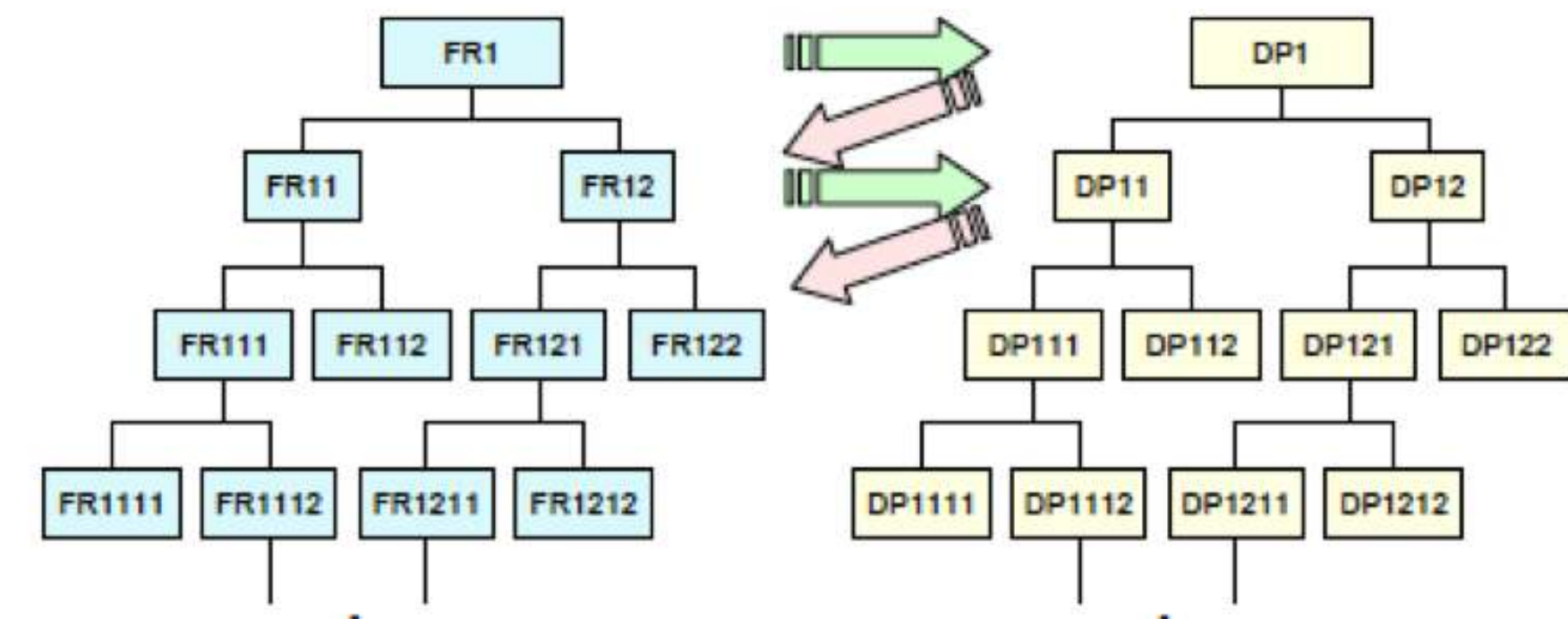


H. Akay, S. Kim, "Reading functional requirements using machine learning-based language processing," CIRP Annals v70.1, 2021

H. Akay, Yang, M., Kim, S-G., 2021, Automatic Design Requirement Extraction from Text with Deep Learning, Proceedings of the American Society of Mechanical Engineers IDETC/CIE, 85390

Design of Agents

- Define functions (Functional Requirements)
- Separate “What’s (FRs)” from “How’s (DPs)”.
- Build "Tree of FRs" as high as possible.
- Repeat to form Tree of Thoughts

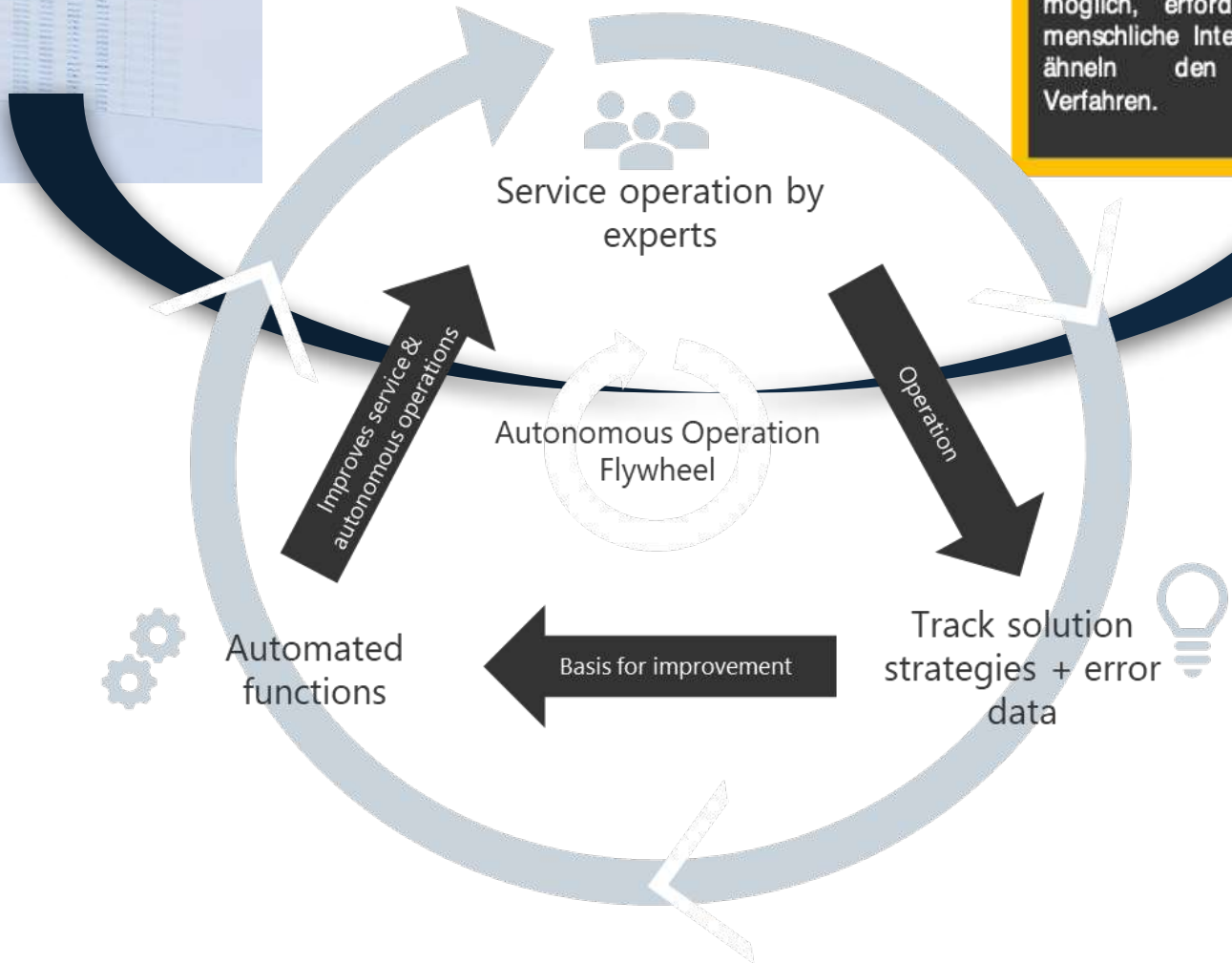


Remote operator as starting point -> Goal: automatic troubleshooting

Today: Remote Operator extensive skills and comprehensive training is needed.



Vision: Machine can eliminate faults itself.



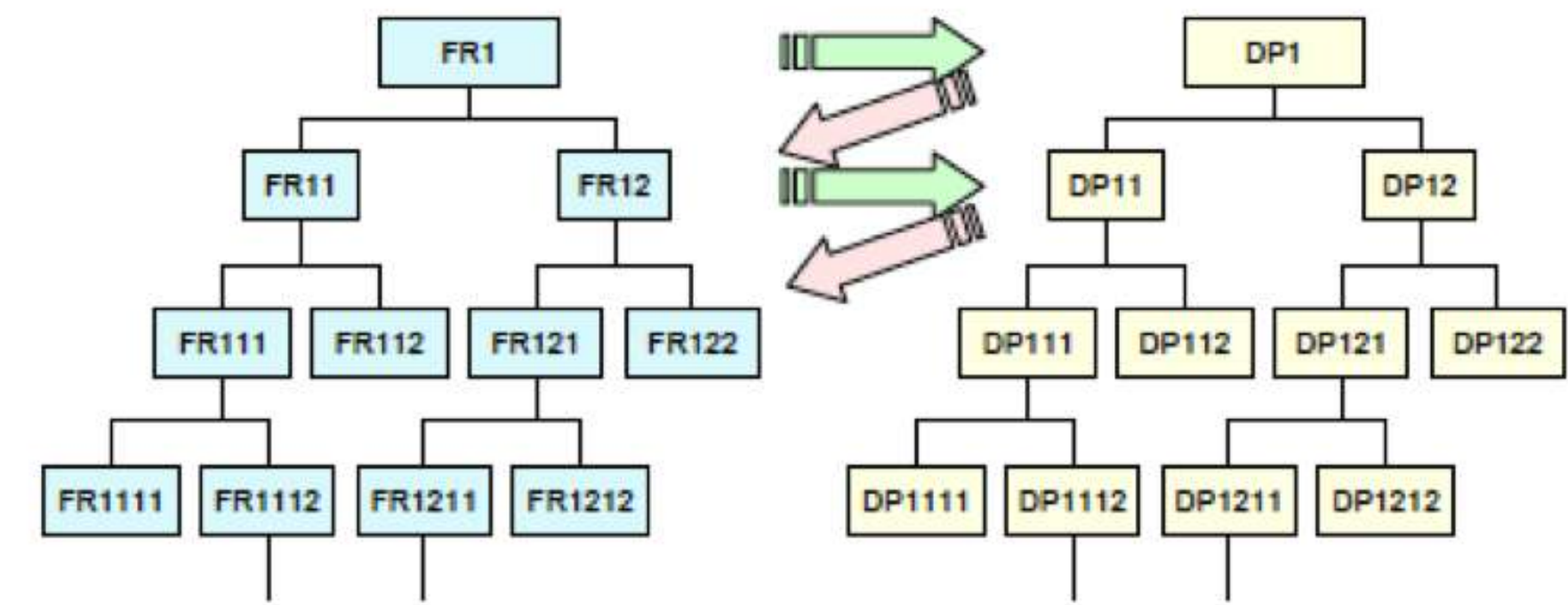
Automatisierung	Industrielle Autonomie	
Remote Operations Kernidee Remote Operator sind für die sichere Durchführung verantwortlich und teilen sich die Arbeitslast mit den Automatisierungssystemen. Rolle des Menschen Überwiegend vor Ort mit einer beträchtlichen Anzahl an manuellen vor-Ort-Operationen. Fernoperationen sind möglich, erfordern jedoch menschliche Interaktion und ähneln den Vor-Ort-Verfahren.	SEMI AUTONOMOUS OPERATION Kernidee Einige Prozesse laufen autonom, menschliches Eingreifen ist jedoch bei Ausnahmen erforderlich. Systeme sind vollständig integriert und koordinieren verschiedene Prozesse autonom. Technologie Zusätzliche Sensoren, Automatisierungen, KI und Edge-Computing unterstützen den Betrieb. Rolle des Menschen Überwachung und Eingreifen bei Bedarf. Nur minimale Eingriffe notwendig.	FULLY AUTONOMOUS OPERATION Kernidee Vollständig autonome und sich selbst optimierende Systeme ohne menschliche Eingriffe. Technologie KI und Automatisierungen ermöglichen eine selbstlernende und optimierende Umgebung. Rolle des Menschen Eingriffe nur bei außergewöhnlichen Situationen.

Confidential

Design of remote troubleshooting agent

Transform what to achieve to Agent submodules

- Define functions (Functional Requirements)
- Separate “What’s (FRs)” from “How’s (DPs)”.
- Build "Tree of FRs" as high as possible.
 1. Buckets and "up & down"
 2. At the highest level, map FR1 to DP1 (or multiple DPs).
 3. No need to specify DPs at the highest level
 4. Then decompose FR1 to FR11 ..FR1n.
 5. Map FR1, .. FR1n to DP1, ..DP1n
- Repeat 1-5 until reach terminal nodes



Machining breakdown

Multi-modal data from real machining operations and breakdowns

System visualization (Real-Data)

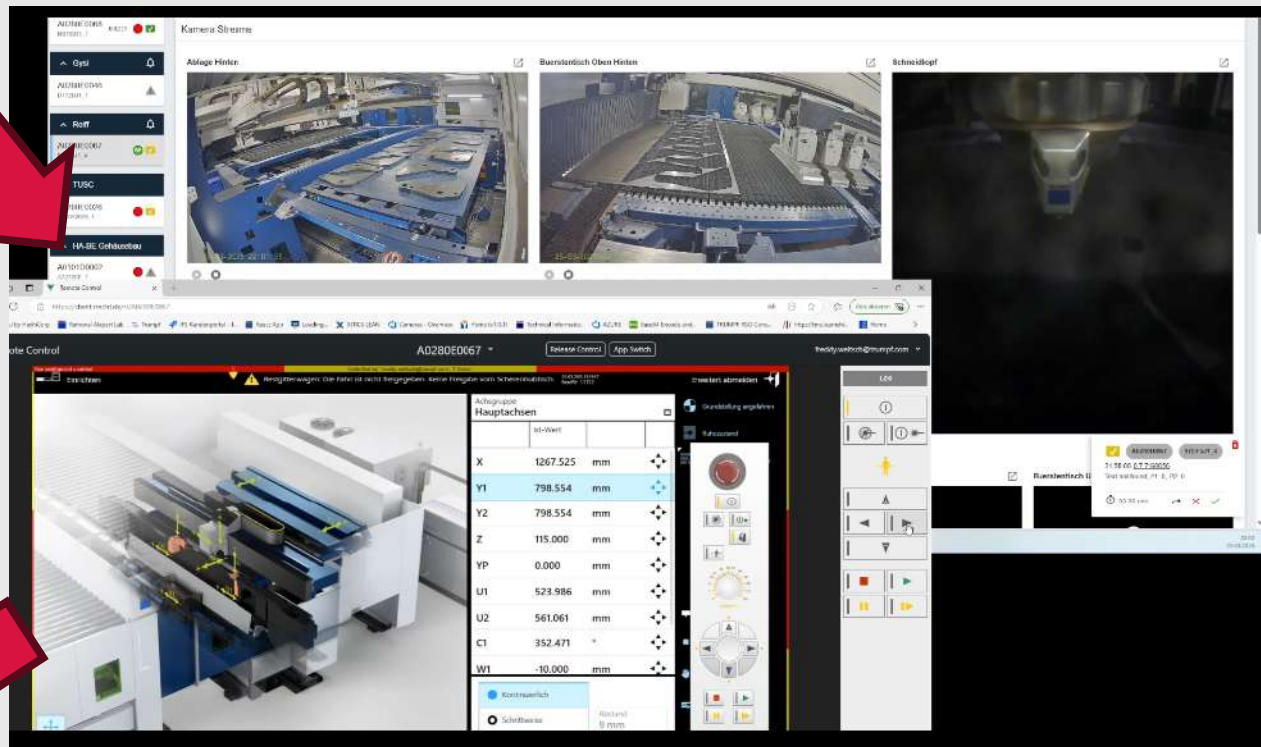
source_timestamp	node_id	value
2025-03-25T20:57:10.362+00:00		179 {"ConditionI
2025-03-25T20:57:12.381+00:00		179 {"ConditionI
2025-03-25T20:57:12.381+00:00		179 {"ConditionI
2025-03-25T20:57:15.367+00:00		179 {"ConditionI
2025-03-25T20:57:15.377+00:00		3039 TRUE
2025-03-25T20:57:15.377+00:00		3037 4
2025-03-25T20:58:08.721+00:00		53 FALSE
2025-03-25T20:58:08.721+00:00		54 TRUE
2025-03-25T20:58:08.721+00:00		59 1160
2025-03-25T20:58:08.731+00:00		3038 1
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2025-03-25T20:58:08.731+00:00		3034 0
2025-03-25T20:58:08.927+00:00	35. StoppedMalfunction	TRUE
2025-03-25T20:58:08.927+00:00	35. Stopped	TRUE
2025-03-25T20:58:08.927+00:00	35. Running	FALSE

Detecting production faults

Input multi-modal data:

- Dynamic real-time videos
- Static knowledge graph
- Machine signals

Multimodal Large Language Model Analysis



Which camera allows for the fastest analysis of faults and derivation of solutions?

Description:

Real machining data:

- 14 Camera angles, PLC data, OPC-UA error codes (100+, 1-2mins around failures, incl. resolution)
- Machining handbook and faults' descriptions .Pdf

> 300.000 tsd.
Videos +
machine data
available for
training

Goal: Adapt autonomously to disturbances during the production

-> Select best camera to view the error + propose resolution strategy based on best practices

Currently underway:

- Static training: PDFs → graph-based knowledge for MLLM pretraining & querying
- Dynamic analysis: Video in OpenCV with VLM/MLLM (e.g., Qwen 2.5)

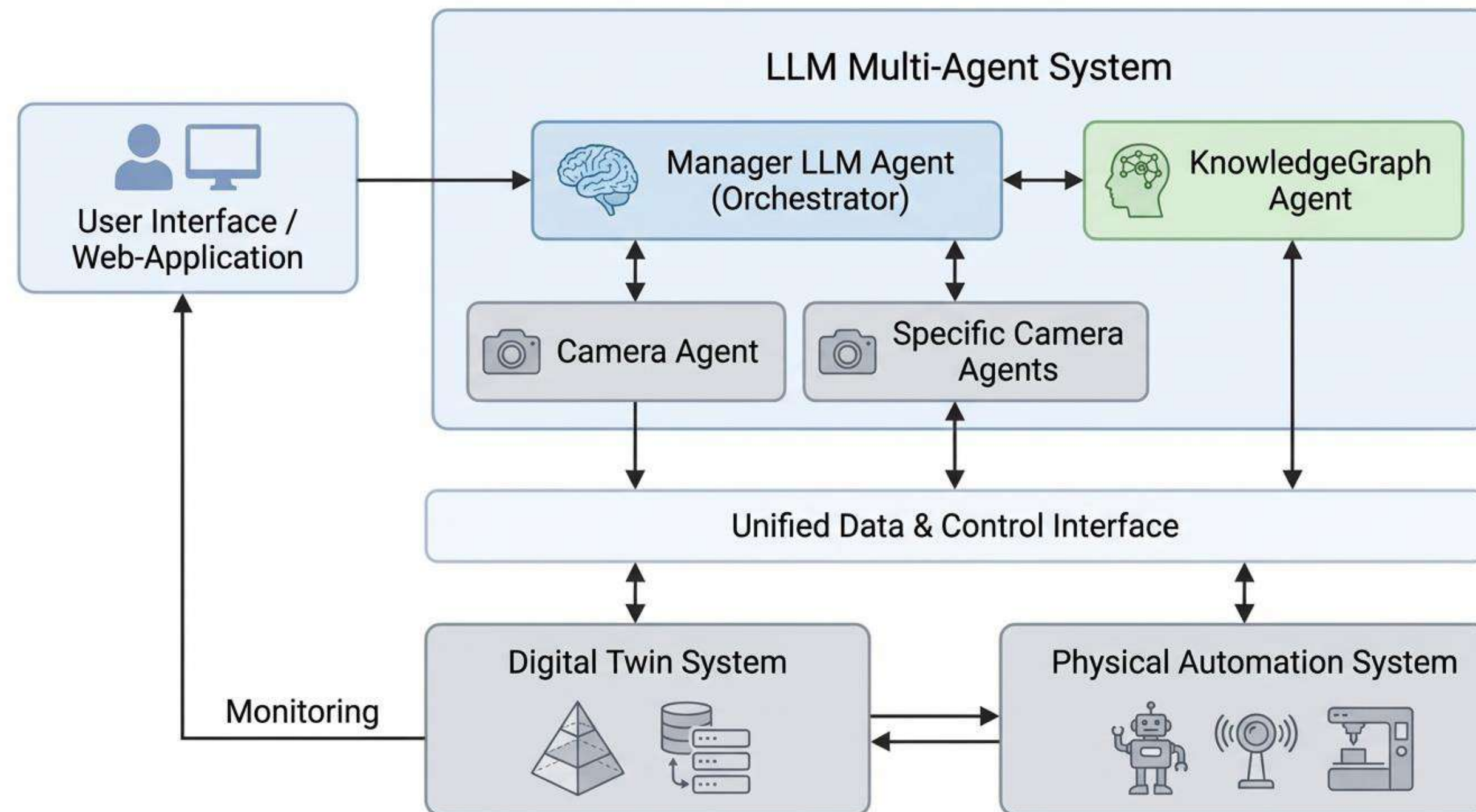
Expected outputs fast-response fault reports:

- Optimal camera view ranking
- Fault cause identification
- Recovery & mitigation strategies

Confidential

Multimodal LLM-VLM Multi Agent Solutions

Physical View



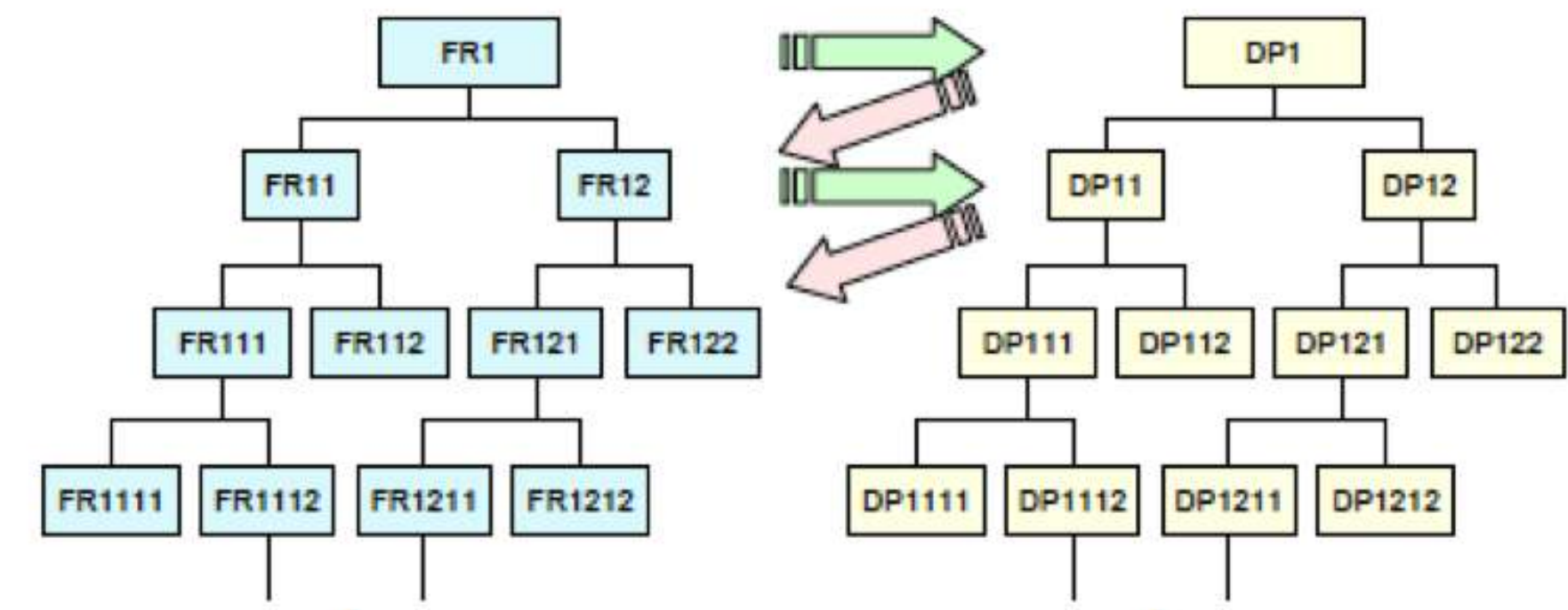
- Is this a functional view of physical view?
- How could we design AI Agents?

Design of remote troubleshooting agent

Functional View

Transform what to achieve to Agent submodules

- Define functions (Functional Requirements)
- Separate “What’s (FRs)” from “How’s (DPs)”.
- Build "Tree of FRs" as high as possible.
 1. Buckets and "up & down"
 2. At the highest level, map FR1 to DP1 (or multiple DPs).
 3. No need to specify DPs at the highest level
 4. Then decompose FR1 to FR11 ..FR1n.
 5. Map FR1, .. FR1n to DP1, ..DP1n
- Repeat 1-5 until reach terminal nodes



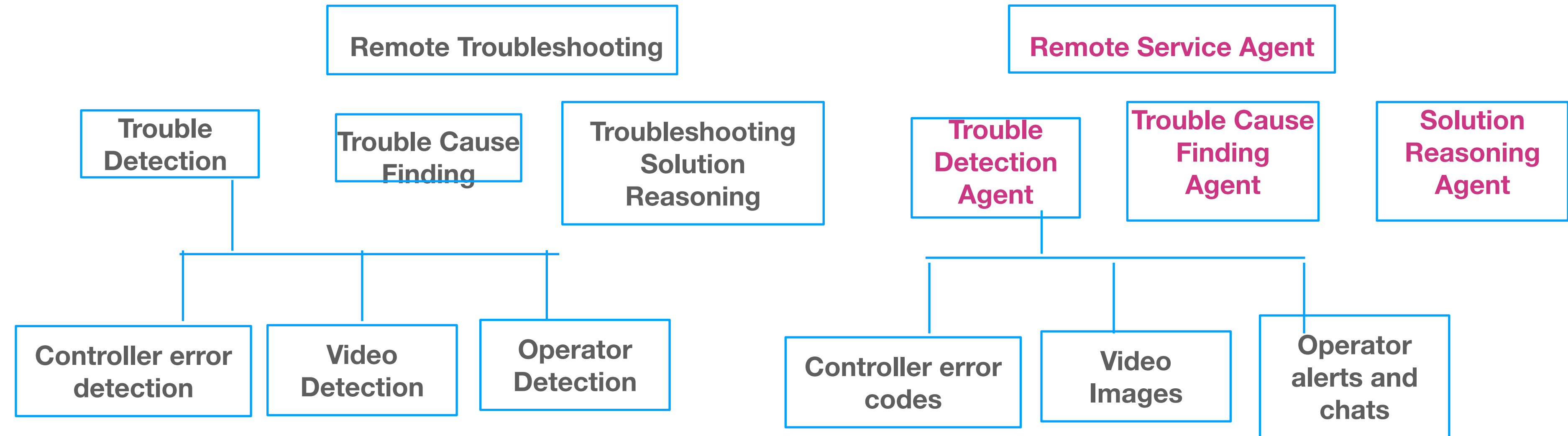
Design of Remote Troubleshooting Agents

Functional View

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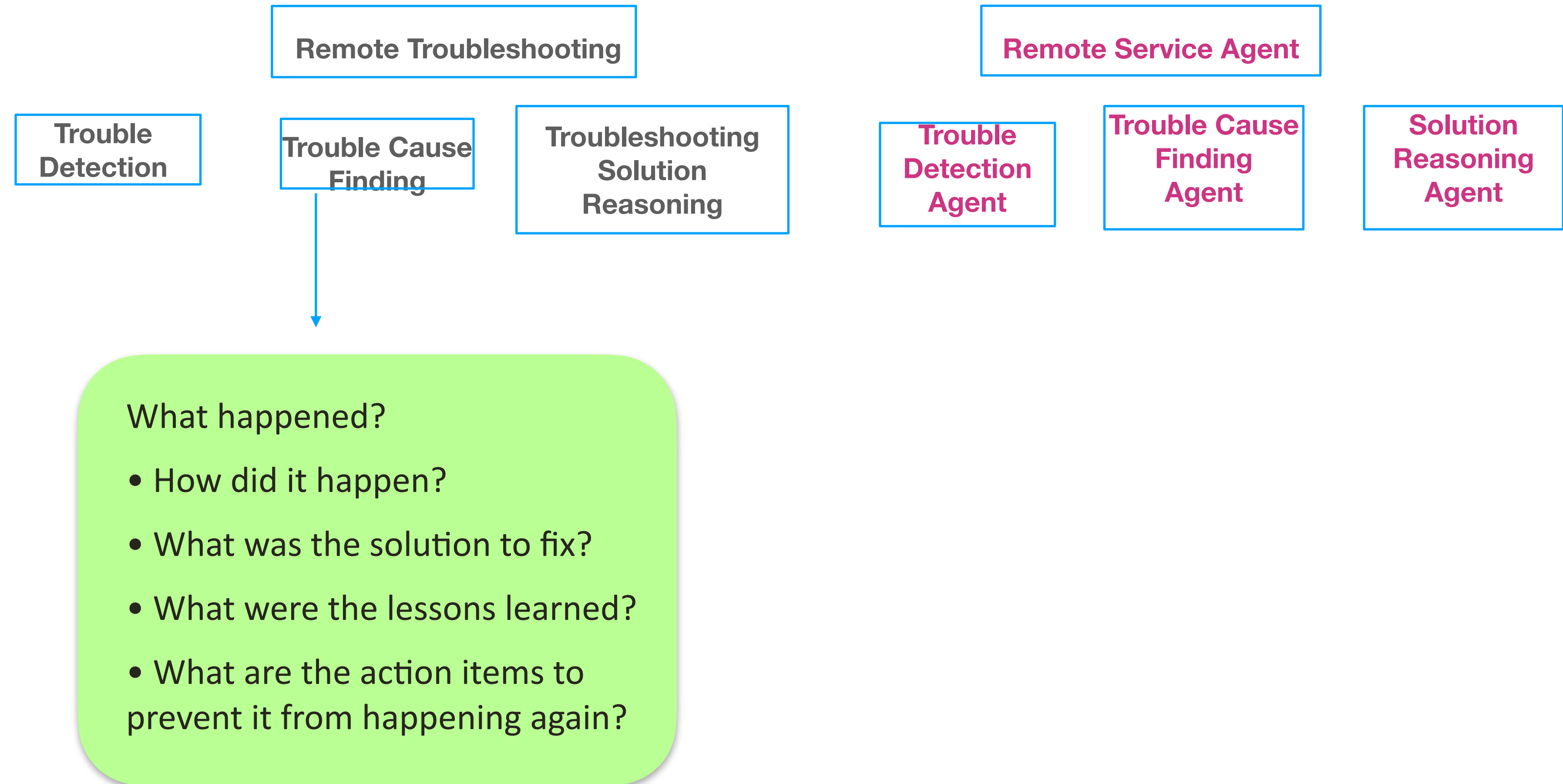
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- Repeat



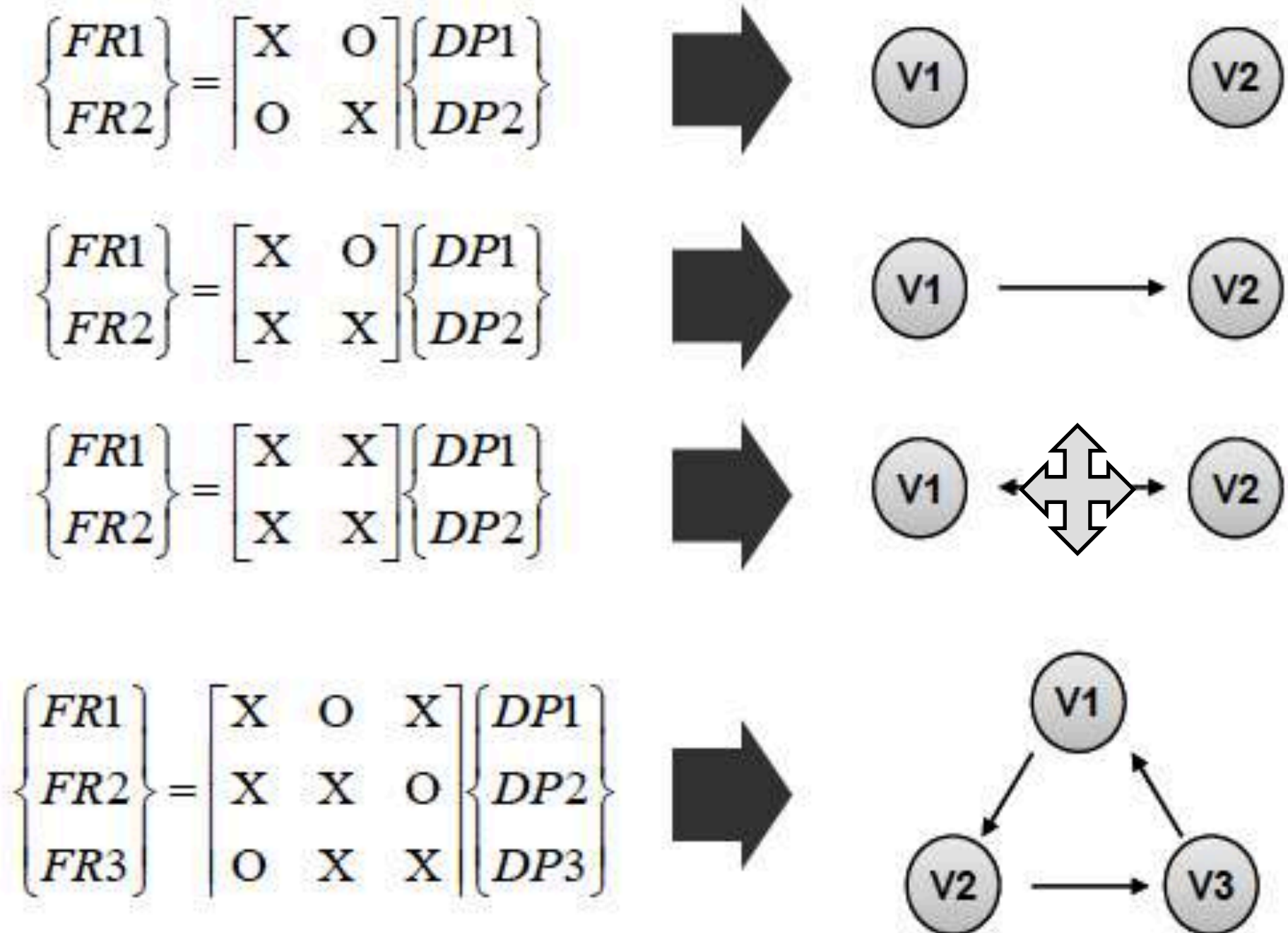
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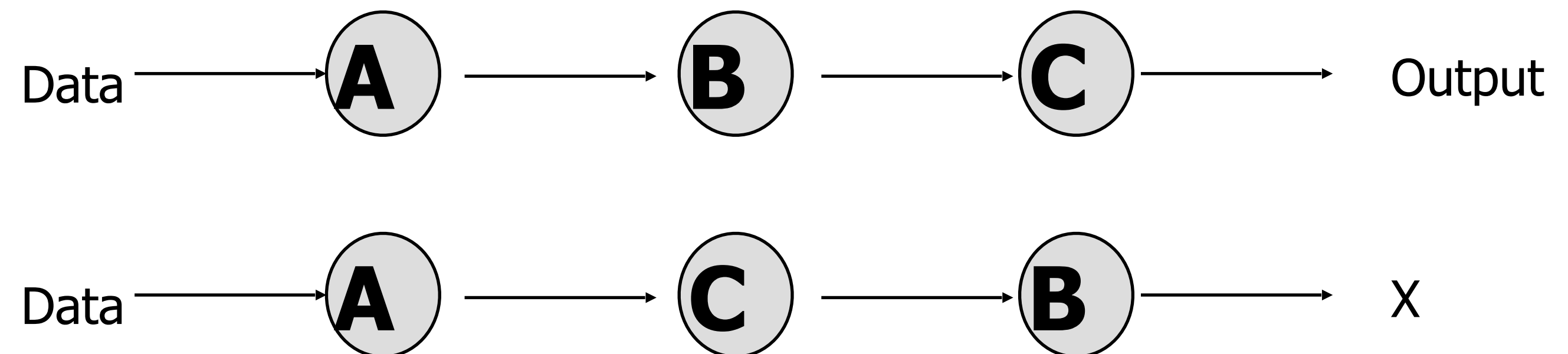
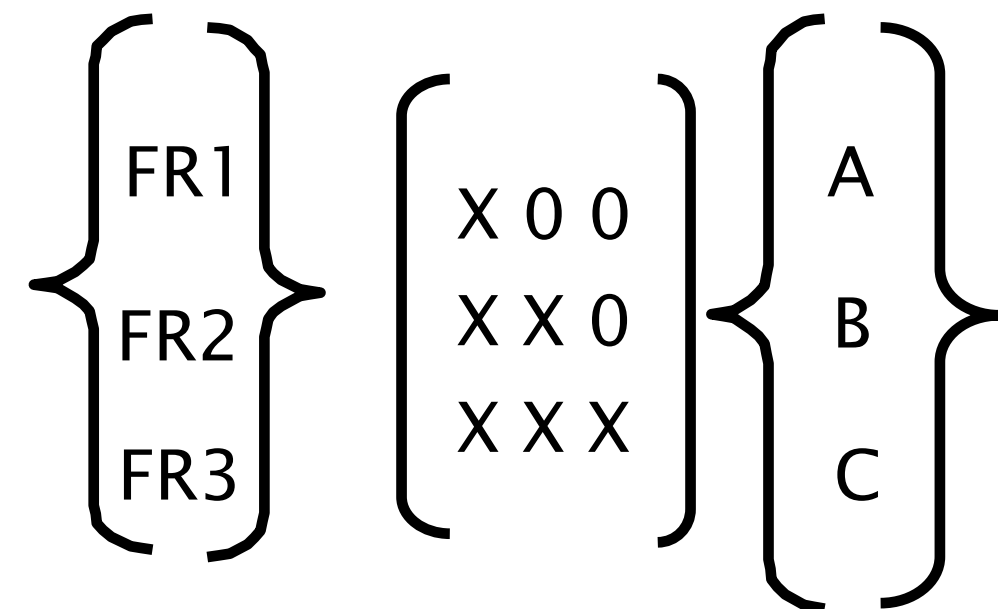
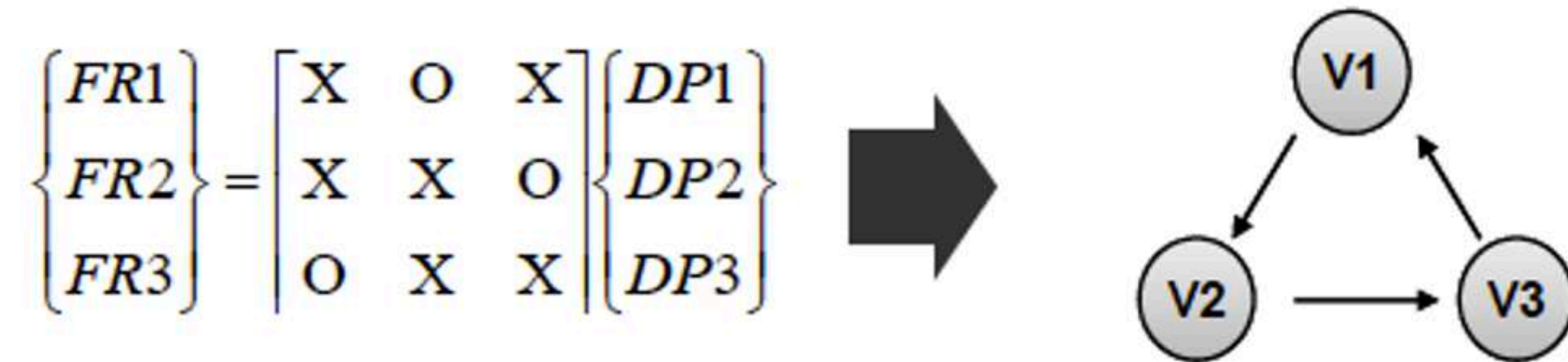


Functional Couplings of Agents

- Orchestration Agent
- Determine execution sequence or decisions
- Follow FR-DP graph
- Make additional queries to find necessary data, information .



Organizational design



Summary:

- Functional thinking is the way to assess the systems design problem.
- Axiomatic approach to design can provide a scientific framework for system architecting.
- System's FR tree (Function Graph) can be the backbone of the digital thread connecting all the activities and decisions of a system.
- True digital transformation will become feasible.

Take Home Messages

- Is Multimodal Modeling a “What” or “How” ?
- Is AI a “What” or “How” ?
- What do you want with AI?
- Systems thinking !!



How to solve large and complex problems?

